

Windows 365 License Management

the Graph API edition



Esther Barthel

Sr. Solutions Architect

Sr. DevOps Engineer



Graph API enthusiast

PowerShell fan



virtues-it.bsky.social



@virtuEs_IT



ebarthel



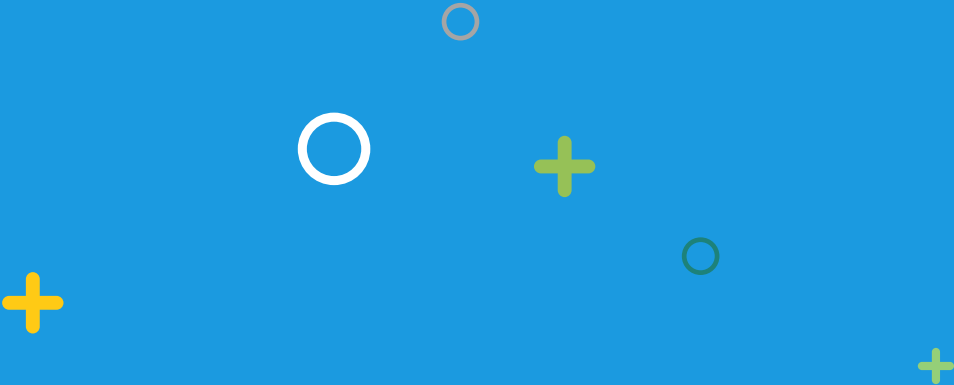
@cloud-devops-ninja



cloud-devops-ninja



www.cloud-devops.ninja



End User Computing



M

C

²

N

M

C

²

M

C

²

plex

N

M C ²



AVD



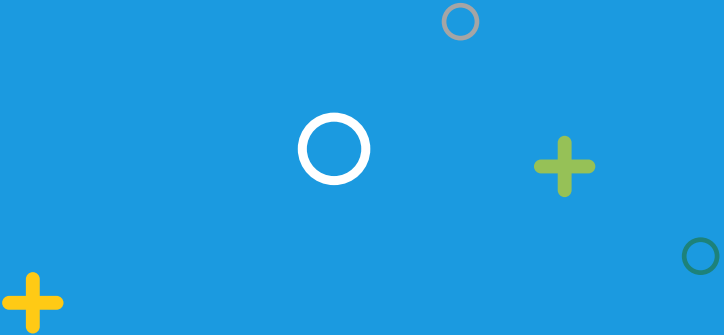
W365 CloudPC



W365 CloudPC



W365 CloudPC



Infrastructure-as-Code



“

*a modern approach to **managing & provisioning**
IT **infrastructure** through the use of **code**,
rather than traditional manual processes.*

”



W365 CloudPC
(Config-as-Code)

Microsoft Graph



W365 CloudPC
(Config-as-Code)

Every Bite a

Better Burger!

Burger Pasta Salads



Beef Burger
Onion With Cheese
\$18.00



Chicken Burger
Cheese With Chicken
\$12.00



Classic Burger
Beef Wit Lottice
\$24.00



Grilled Burger
Griller Chicken
\$14.00



Classic Burger
Beef Wit Lottice
\$24.00

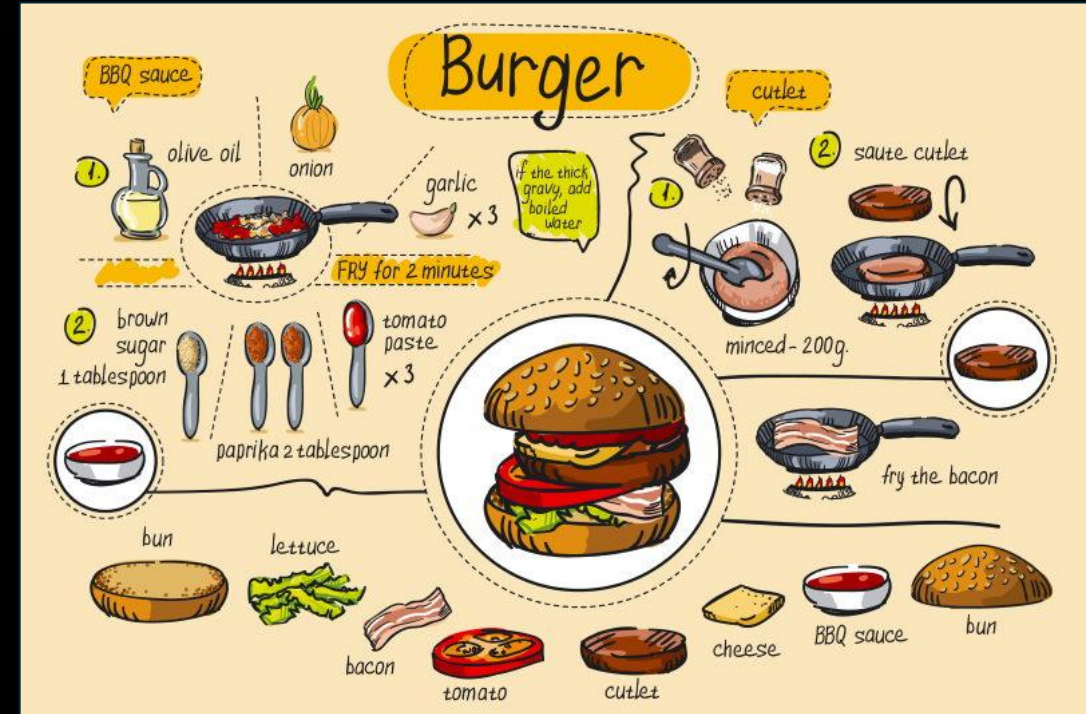


Grilled Burger
Griller Chicken
\$14.00



**Fresh Beef
Burger With Cheese...**

Beef burger With Fresh Cheese And Onion Served With Fries And Drink. Enjoy our 20% Off With New Promo Code

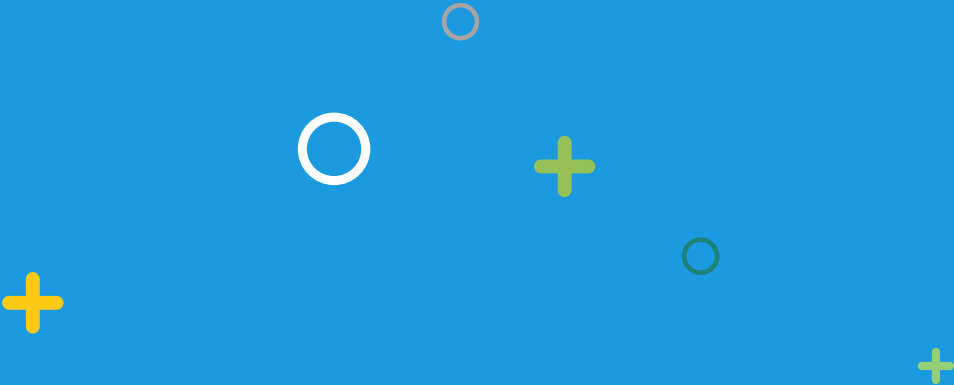


Declarative code
(templates)



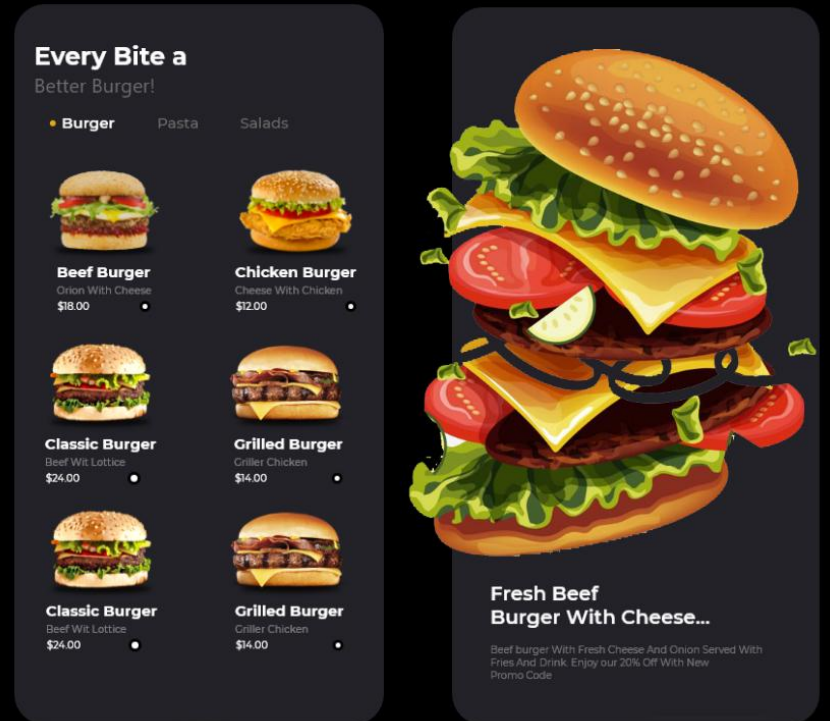
Imperative code
(scripts)





Getting Started with the Graph API





Declarative code (templates)



Microsoft
Graph



Declarative code (templates)

Every Bite a
Better Burger!

Burger

Pasta

Salads



Beef Burger
Onion With Cheese
\$18.00



Chicken Burger
Cheese With Chicken
\$12.00



Classic Burger
Beef With Lettuce
\$24.00



Grilled Burger
Griller Chicken
\$14.00



Classic Burger
Beef With Lettuce
\$24.00



Grilled Burger
Griller Chicken
\$14.00



**Fresh Beef
Burger With Cheese...**

Beef burger With Fresh Cheese And Onion Served With
Fries And Drink. Enjoy our 20% Off With New
Promo Code



```
main-group.bicep > ...
1 extension 'br:mcr.microsoft.com/bicep/extensions/microsoftgraph/v1.0:0.1.8-preview'
2
3 param location string = resourceGroup().location
4 param groupName string = 'myUniqueGroup'
5
6 resource managedIdentity 'Microsoft.ManagedIdentity/userAssignedIdentities@2023-01-31' = {
7   name: 'ExampleManagedIdentity20240805'
8   location: location
9 }
10
11
12
```



Azure Resource
Manager

Deployments
Engine

Azure
resources

Microsoft Graph
resources

Microsoft Graph
Bicep extension



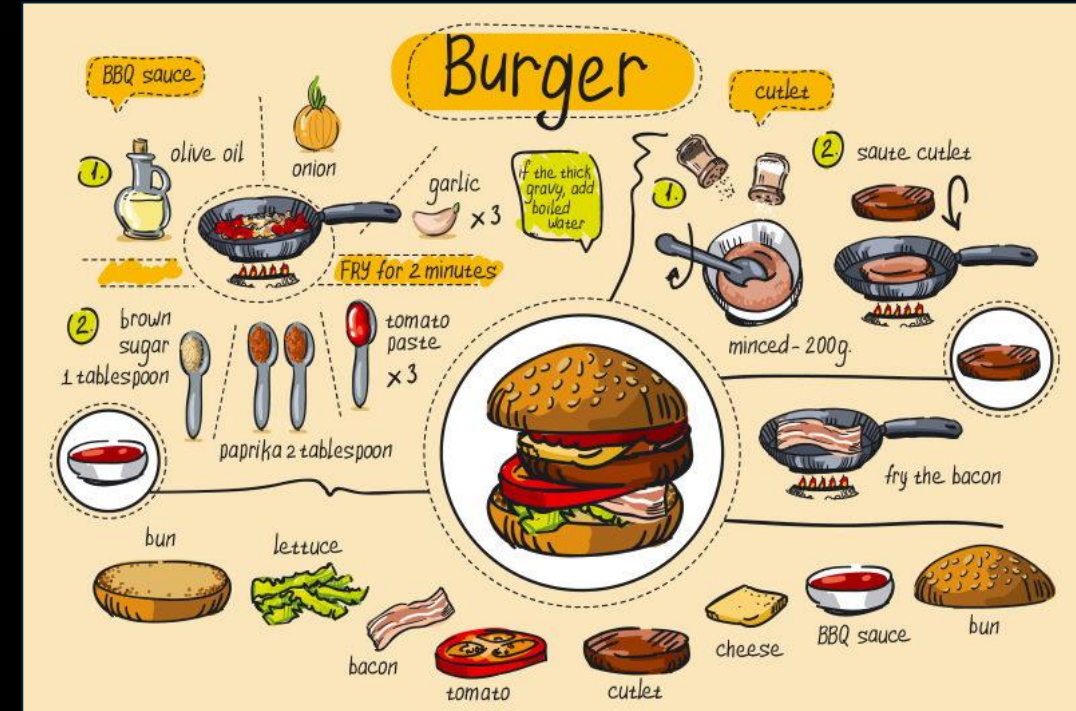
Microsoft
Graph

Microsoft Graph Bicep v1.0 resource reference overview

This article gives you an overview of the Microsoft Graph resource types you can use in Bicep files, for the `v1.0` endpoint.

Supported resources

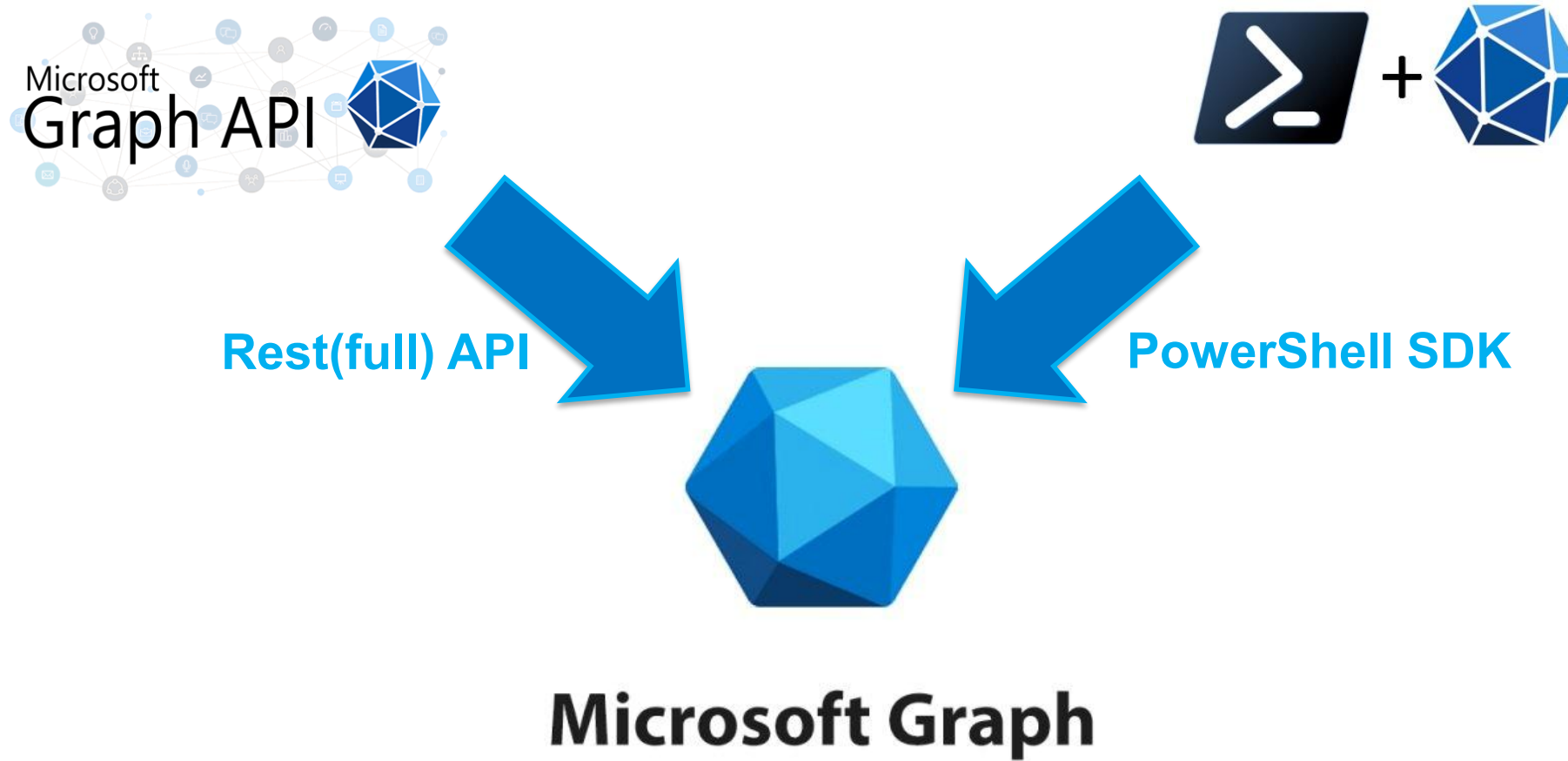
Service	Microsoft Graph Bicep resource type
Applications	<code>Microsoft.Graph/applications</code>
App role assignments	<code>Microsoft.Graph/appRoleAssignedTo</code>
Federated identity credentials	<code>Microsoft.Graph/applications/federatedIdentityCredentials</code>
Groups	<code>Microsoft.Graph/groups</code>
OAuth2 permission grants (delegated permission grants)	<code>Microsoft.Graph/oauth2PermissionGrants</code>
Service principals	<code>Microsoft.Graph/servicePrincipals</code>
Users	<code>Microsoft.Graph/users</code>



Imperative code (scripts)



Coding skills loading





PSCloudPC - Windows 365 Management PowerShell Module

[Home](#) [Getting started](#) [User Guide](#) [Contributing](#) [About](#)

Windows 365 Cloud PC Management PowerShell Module

Description

This PowerShell module allows you to manage your Windows 365 environment from the command line. It provides a set of cmdlets that allow you to perform various tasks, such as creating, modifying and deleting policies, managing Cloud PCs, and more.

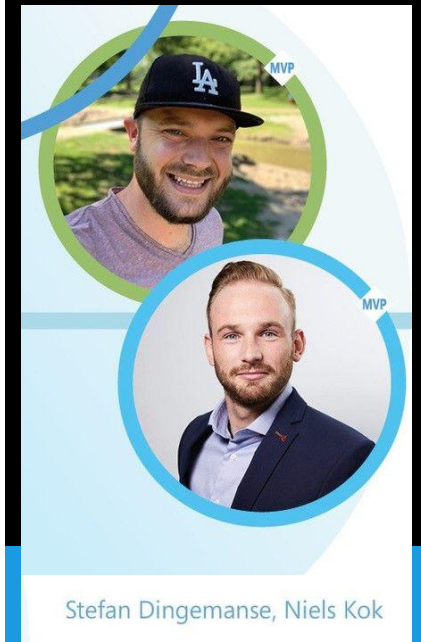


Getting Started

```
Install-Module -Name PScloudPC -Verbose
```

Then import the module into your session

```
Import-Module PScloudPC -Verbose -Force
```



Stefan Dingemanse, Niels Kok

Graph Explorer

Sample queries

Resources

Search resources

Switch to beta ☐ Off

Resources available

groups (8)

GET

POST

> {group-id} (46)

> \$count (1)

GET

v1.0

`https://graph.microsoft.com/v1.0/groups?$filter=startswith(displayName, 'grp-sec')&$select=id,displayName,description`



Run query

Request body

Request headers

Modify permissions

Access token

OK - 200 - 333 ms



Response preview

Response headers

Code snippets

Toolkit component

Adaptive cards

```
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#groups(id,displayName,description)",
  "value": [
    {
      "id": "[REDACTED]",
      "displayName": "grp-sec-AVDUsers",
      "description": "AVD Users group"
    }
  ]
}
```

Response preview

Response headers

Code snippets

Toolkit component

Adaptive cards

C#

CLI

Go

Java

JavaScript

PHP

PowerShell

Python

Leverage the power of our client libraries. Download the PowerShell client library here <https://aka.ms/pshellsdk>
To read more about our SDKs, go to the documentation page at <https://aka.ms/pshellsdkdocs>

Import-Module Microsoft.Graph.Groups

Get-MgGroup -Filter "startswith(displayName, 'grp-sec')" -Property "id,displayName,description"

Graph API reference docs



Use the Microsoft Graph API

Microsoft Graph is a RESTful web API that enables you to access Microsoft Cloud service resources. After you [register your app](#) and [get authentication tokens for a user or service](#), you can make requests to the Microsoft Graph API.

Call a REST API method

To read from or write to a resource such as a user or an email message, you construct a request that looks like the following:

HTTP

 Copy

```
{HTTP method} https://graph.microsoft.com/{version}/{resource}?{query-parameters}
```

4. Request an access token

In the OAuth 2.0 client credentials grant flow, you use the application ID and client secret values that you saved when you registered your app to request an access token directly from the Microsoft identity platform `/token` endpoint.

You specify the preconfigured permissions by passing `https://graph.microsoft.com/.default` as the value for the `scope` parameter in the token request.

Send a POST request to the `/token` identity platform endpoint to acquire an access token. In this request, the client uses the client secret.

HTTP

cURL

HTTP

 Copy

request, the client uses the client secret.

HTTP

cURL

HTTP

 Copy

// Line breaks are for legibility only.

POST https://login.microsoftonline.com/{tenant}/oauth2/v2.0/token HTTP/1.1

Host: login.microsoftonline.com

Content-Type: application/x-www-form-urlencoded

client_id=535fb089-9ff3-47b6-9bfb-4f1264799865

&scope=https%3A%2F%2Fgraph.microsoft.com%2F.default

&client_secret=qWgdYA...L1qKv5bPX

&grant_type=client_credentials

request, the client uses the client secret.

HTTP

cURL

HTTP

 Copy

// Line breaks are for legibility only.

POST <https://login.microsoftonline.com/{tenant}/oauth2/v2.0/token> HTTP/1.1

Host: login.microsoftonline.com

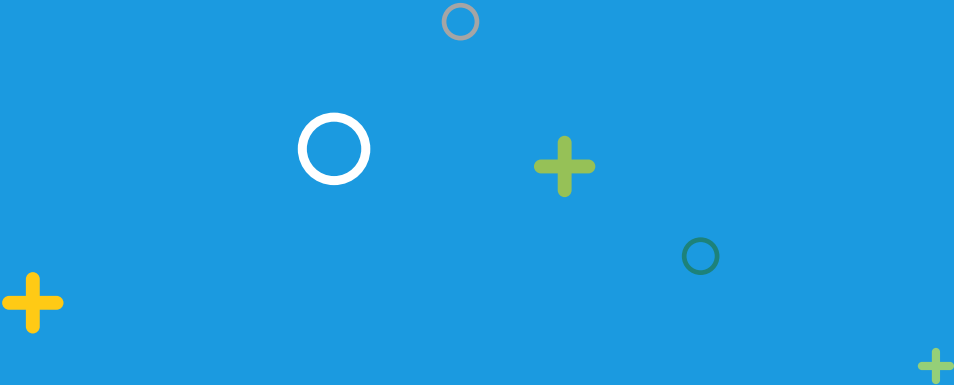
Content-Type: application/x-www-form-urlencoded

client_id=535fb089-9ff3-47b6-9bfb-4f1264799865

&scope=https%3A%2F%2Fgraph.microsoft.com%2F.default

&client_secret=qWgdYA...L1qKv5bPX

&grant_type=client_credentials



IT Processes



W365 License Management

explaining the workflow



W365 CloudPC License Management

M

C

2

Get-W365CPC-InactiveFor60Days-Report-Data (phase 1)

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

Revoke-W365CPC-License (phase 2)

Remove User from
CloudPC license group
(Enterprise CPC)

Remove User from
CloudPC provisioning group
(Enterprise & Frontline CPC)

Notify-W365CPC-User (phase 3)

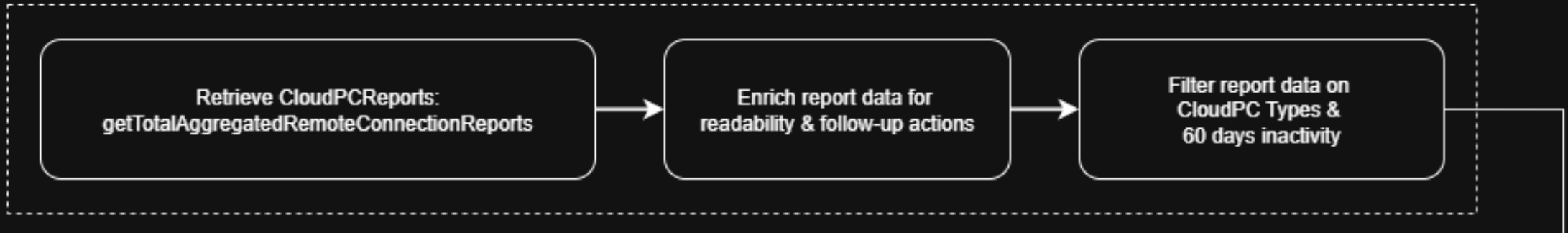
Send notification email
to CloudPC User

M

W365 CloudPC License Management



Get-W365CPC-InactiveFor60Days-Report-Data (phase 1)



W365 CloudPC License Management



HTTP request

HTTP Copy

POST /deviceManagement/virtualEndpoint/reports/getTotalAggregatedRemoteConnectionReports

Request headers

Expand table

Name	Description
Authorization	Bearer {token}. Required. Learn more about authentication and authorization.
Content-Type	application/json. Required.

Request body

In the request body, supply a JSON representation of the parameters.

The following table shows the parameters that can be used with this action.

Expand table

Parameter	Type	Description
filter	String	OData filter syntax. Supported filters include <code>and</code> , <code>or</code> , <code>it</code> , <code>le</code> , <code>gt</code> , <code>ge</code> , and <code>eq</code> .
select	String collection	OData select syntax. Represents the selected columns of the reports.
search	String	Specifies a string to search
groupBy	String collection	Specify how to group the reports. If used, must have the same contents as select parameter
orderBy	String collection	Specify how to sort the reports.
skip	Int32	Number of records to skip.
top	Int32	The number of top records to return.



W365 License Management

Postman API testing



W365 CloudPC License Management (Postman)

The screenshot displays the Postman application interface. On the left sidebar, the 'My Workspace' section is active, showing a collection of API requests under the '03. License Management' folder. The main panel shows a POST request to the endpoint `https://graph.microsoft.com/beta/deviceManagement/virtualEndpoint/reports/getTotalAggregatedRemoteConnectionReports/?$count=1...`. The request body is a JSON object with the following structure:

```
{
  "skip": 0,
  "select": [],
  "top": 100,
  "search": "",
  "orderBy": [
    "TotalUsageInHour"
  ]
}
```

The response is a 200 OK status, indicating a successful request. The response body is a JSON object with the following structure:

```
{
  "TotalRowCount": 2,
  "Schema": [
    ...
  ],
  "Values": [
    [
      ...
    ],
    [
      ...
    ],
    [
      "627a21fe-a554-47ca-9a3c-f14fe8c3e99b",
      "CPC-test0-N61MG",
      "test01@cognitionitdev.onmicrosoft.com",
      null,
      0.0,
      "645f6084-98ba-4ed4-9299-817ed164860d",
      ...
    ]
  ]
}
```

The bottom status bar shows the response details: 200 OK, 1.07 s, 2.03 KB. The bottom right corner contains various utility icons: Postbot, Runner, Capture requests, Cloud Agent, Cookies, Vault, and Trash.

W365 CloudPC License Management (Postman)

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

HTTP 00. GraphAPI Demos / Windo... / 03. License Management / CloudPCReports: getTotalAggregatedRemoteConnectionReports Save Share

POST `https://graph.microsoft.com/beta/deviceManagement/virtualEndpoint/reports/getTotalAggregatedRemoteConnectionReports/?$count=1...` Send

Params Authorization Headers (12) **Body** Scripts Settings Cookies Beautify

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON**

```
1 {
2   "skip": 0,
3   "select": [],
4   "top": 1,
5   "search": "",
6   "orderBy": [
7     "TotalUsageInHour"
8   ]
9 }
```

Body Cookies Headers (7) Test Results 200 OK • 783 ms • 1.64 KB Save Response

JSON Visualize

```
1 {
2   "TotalRowCount": 2,
3   "Schema": [ ...
68 ],
69   "Values": [ ...
88 ]
89 }
```

W365 CloudPC License Management (Postman)



00. GraphAPI Demos / Windo... / 03. License Management / CloudPCReports: getTotalAggregatedRemoteConnectionReports

POST

https://graph.microsoft.com/beta/deviceManagement/virtualEndpoint/reports/getTotalAggregatedRemoteConnectionReports/?\$count=1...

Send

Params

Authorization

Headers (12)

Body

Scripts

Settings

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

```
1 {
2   "skip": 0,
3   "select": [],
4   "top": 1,
5   "search": "",
6   "orderBy": [
7     "TotalUsageInHour"
8   ]
9 }
```

Body

Cookies

Headers (7)

Test Results

200 OK

783 ms

1.64 KB

Save Response

JSON

Visualize

```
1 {
2   "TotalRowCount": 2,
3   "Schema": [ ...
```

ManagedDeviceName	DeviceType	NeverSignedIn	LastActiveTime	CreatedDate	DaysSinceLastSignIn	TotalUsageInHour	UserPrincipalName	IntuneDeviceId	PolicyName	ServicePlanId	CloudPCSize	CloudPcId
TST-test0-B5JM4	Enterprise	TRUE		2025-12-05 12:07			0 test0@cognitionittst.onmicrosoft.com	12345678-9012-3 TST Test Cloud PC		2de9c682-1234-1234-1234-1234567890	Cloud PC Enterprise 4vCPU/16GB/128GB	64379843-1234
TST-test1-KP47C	Enterprise	FALSE	6/20/2025 2:37:02 PM	2025-12-05 12:07	56		0 test1@cognitionittst.onmicrosoft.com	12345678-9012-3 TST Test Cloud PC		69dc175c-1234-1234-1234-1234567890	Cloud PC Enterprise 8vCPU/256GB/256GB	cad55617-1234
DEV-test0-VEOL3	Enterprise	FALSE	7/16/2025 9:24:11 AM	7/14/2023 9:37:07 AM	30		0 test0@cognitionitdev.onmicrosoft.com	12345678-9012-3 DEV Development Cloud PC		2de9c682-1234-1234-1234-1234567890	Cloud PC Enterprise 4vCPU/16GB/128GB	82b02dbd-1234
DEV-test1-ZSSX8	FrontlineDedicated	FALSE	7/16/2025 5:55:31 PM	2025-09-07 08:38	30		0 test1@cognitionitdev.onmicrosoft.com	12345678-9012-3 DEV Development Cloud PC FL		057efbfe-1234-1234-1234-12345678901	Cloud PC Frontline 2vCPU/8GB/128GB	6a4f5c5d-1234
DEV-test2-FAUF8	Enterprise	FALSE	6/30/2025 7:09:21 AM	2024-08-04 06:28	46		0 test2@cognitionitdev.onmicrosoft.com	12345678-9012-3 DEV Development Cloud PC		3efff3fe-1234-1234-1234-12345678901	Cloud PC Enterprise 2vCPU/8GB/128GB	25b0a8c9-1234
DEV-test3-D1U44	FrontlineDedicated	TRUE		8/13/2025 8:08:52 AM			0 test3@cognitionitdev.onmicrosoft.com	12345678-9012-3 DEV Development Cloud PC FL		057efbfe-1234-1234-1234-12345678901	Cloud PC Frontline 2vCPU/8GB/128GB	ccceb673-1234
PRD-test1-01387	FrontlineDedicated	TRUE		2025-06-05 07:45			0 test1@cognitionit.com	12345678-9012-3 PRD Production Cloud PC		057efbfe-1234-1234-1234-12345678901	Cloud PC Frontline 2vCPU/8GB/128GB	a65f5965-1234
PRD-test2-0EU22	FrontlineDedicated	FALSE	2025-11-07 12:57	11/14/2024 10:04:12 AM	35		0 test2@cognitionit.com	12345678-9012-3 PRD Production Cloud PC		057efbfe-1234-1234-1234-12345678901	Cloud PC Frontline 2vCPU/8GB/128GB	1d044521-1234

W365 CloudPC License Management (Postman)

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

HTTP

00. GraphAPI Demos / Windo... / 03. License Management / CloudPCReports: getTotalAggregatedRemoteConnectionReports

Save

Share

POST

https://graph.microsoft.com/beta/deviceManagement/virtualEndpoint/reports/getTotalAggregatedRemoteConnectionReports/?\$count=1...

Send

Params

Authorization

Headers (12)

Body

Scripts

Settings

Cookies

Beautify

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

```
1 {
2   "skip": 0,
3   "select": [],
4   "top": 1,
5   "search": "",
6   "orderBy": [
7     "TotalUsageInHour"
8   ]
9 }
```

Body

Cookies

Headers (7)

Test Results

200 OK

783 ms

1.64 KB

Save Response

JSON

Visualize

```
1 {
2   "TotalRowCount": 2,
3   "Schema": [ ...
```

ManagedDeviceName	DeviceType	NeverSignedIn	LastActiveTime	CreatedDate	DaysSinceLastSignIn	TotalUsageInHour	UserPrincipalName	IntuneDeviceId	PolicyName	ServicePlanId	CloudPCSize	CloudPcId
TST-test0-B5JM4	Enterprise	TRUE		2025-12-05 12:07			test0@cognitionittst.onmicrosoft.com	12345678-9012-3456-7890-123456789012	TST Test Cloud PC	2de9c682-1234-1234-1234-123456789012	Cloud PC Enterprise 4vCPU/16GB/128GB	64379843-1234-1234-1234-123456789012
TST-test1-KP47C	Enterprise	FALSE	6/20/2025 2:37:02 PM	2025-12-05 12:07	56		test1@cognitionittst.onmicrosoft.com	12345678-9012-3456-7890-123456789012	TST Test Cloud PC	69dc175c-1234-1234-1234-123456789012	Cloud PC Enterprise 8vCPU/256GB/256GB	cad55617-1234-1234-1234-123456789012
DEV-test0-VEOL3	Enterprise	FALSE	7/16/2025 9:24:11 AM	7/14/2023 9:37:07 AM	30		test0@cognitionitdev.onmicrosoft.com	12345678-9012-3456-7890-123456789012	DEV Development Cloud PC	2de9c682-1234-1234-1234-123456789012	Cloud PC Enterprise 4vCPU/16GB/128GB	82b02dbd-1234-1234-1234-123456789012
DEV-test1-ZSSX8	FrontlineDedicated	FALSE	7/16/2025 5:55:31 PM	2025-09-07 08:38	30		test1@cognitionitdev.onmicrosoft.com	12345678-9012-3456-7890-123456789012	DEV Development Cloud PC FL	057efbfe-1234-1234-1234-123456789012	Cloud PC Frontline 2vCPU/8GB/128GB	6a4f5c5d-1234-1234-1234-123456789012
DEV-test2-FAUF8	Enterprise	FALSE	6/30/2025 7:09:21 AM	2024-08-04 06:28	46		test2@cognitionitdev.onmicrosoft.com	12345678-9012-3456-7890-123456789012	DEV Development Cloud PC	3efff3fe-1234-1234-1234-123456789012	Cloud PC Enterprise 2vCPU/8GB/128GB	25b0a8c9-1234-1234-1234-123456789012
DEV-test3-D1U44	FrontlineDedicated	TRUE		8/13/2025 8:08:52 AM			test3@cognitionitdev.onmicrosoft.com	12345678-9012-3456-7890-123456789012	DEV Development Cloud PC FL	057efbfe-1234-1234-1234-123456789012	Cloud PC Frontline 2vCPU/8GB/128GB	ccceb673-1234-1234-1234-123456789012
PRD-test1-01387	FrontlineDedicated	TRUE		2025-06-05 07:45			test1@cognitionit.com	12345678-9012-3456-7890-123456789012	PRD Production Cloud PC	057efbfe-1234-1234-1234-123456789012	Cloud PC Frontline 2vCPU/8GB/128GB	a65f5965-1234-1234-1234-123456789012
PRD-test2-0EU22	FrontlineDedicated	FALSE	2025-11-07 12:57	11/14/2024 10:04:12 AM	35		test2@cognitionit.com	12345678-9012-3456-7890-123456789012	PRD Production Cloud PC	057efbfe-1234-1234-1234-123456789012	Cloud PC Frontline 2vCPU/8GB/128GB	1d044521-1234-1234-1234-123456789012

W365 CloudPC License Management (Postman)

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

00. GraphAPI Demos / Windo... / 03. License Management / CloudPCReports: getTotalAggregatedRemoteConnectionReports

POST https://graph.microsoft.com/beta/deviceManagement/virtualEndpoint/reports/getTotalAggregatedRemoteConnectionReports/?\$count=1... Send

Params Authorization Headers (12) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "skip": 0,
3   "select": [],
4   "top": 1,
5   "search": "",
6   "orderBy": [
7     "TotalUsageInHour"
8   ]
9 }
```

Body Cookies Headers (7) Test Results 200 OK • 783 ms • 1.64 KB Save Response

{ JSON Visualize

```
1 {
2   "TotalRowCount": 2,
3   "Schema": [ ...
```

ManagedDeviceName	DeviceType	NeverSignedIn	LastActiveTime	CreatedDate	DaysSinceLastSignIn	UserPrincipalName	contactEmail	ServicePlanId	ServicePlanName	provisioningPolicyGroupId	provisioningPolicyGroupName	ExpectedEnterpriseLicenseGroupName
TST-test0-B5JM4	Enterprise	TRUE		2025-12-05 12:07		test0@cognitionittst.onmicro	test0@cognitionit.com	2de9c682-1234	Cloud PC Enterprise 4vCPU/16GB/128GB	736b9fc8-1234-1234-1234	W365-TST Test Cloud PC	W365-LIC-4vCPU-16GB-128GB
TST-test1-KP47C	Enterprise	FALSE	6/20/2025 2:37:02 PM	2025-12-05 12:07	56	test1@cognitionittst.onmicro	test1@cognitionit.com	69dc175c-1234	Cloud PC Enterprise 8vCPU/32GB/256GB	736b9fc8-1234-1234-1234	W365-TST Test Cloud PC	W365-LIC-8vCPU-32GB-256GB
DEV-test0-VEOL3	Enterprise	FALSE	7/16/2025 9:24:11 AM	7/14/2023 9:37:07 AM	30	test0@cognitionitdev.onmicrc	test0@cognitionit.com	2de9c682-1234	Cloud PC Enterprise 4vCPU/16GB/128GB	b8483202-1234-1234-1234	W365-DEV Development Cloud PC	W365-LIC-4vCPU-16GB-128GB
DEV-test1-ZSSX8	FrontlineDedicated	FALSE	7/16/2025 5:55:31 PM	2025-09-07 08:38	30	test1@cognitionitdev.onmicrc	test1@cognitionit.com	057efbfe-1234	Cloud PC Frontline 2vCPU/8GB/128GB	a0249b00-1234-1234-1234	W365-DEV Development Cloud PC FL	
DEV-test2-FAUF8	Enterprise	FALSE	6/30/2025 7:09:21 AM	2024-08-04 06:28	46	test2@cognitionitdev.onmicrc	test2@cognitionit.com	3efff3fe-1234-1	Cloud PC Enterprise 2vCPU/8GB/128GB	b8483202-1234-1234-1234	W365-DEV Development Cloud PC	W365-LIC-2vCPU-8GB-128GB
DEV-test3-D1U44	FrontlineDedicated	TRUE		8/13/2025 8:08:52 AM		test3@cognitionitdev.onmicrc	test3@cognitionit.com	057efbfe-1234	Cloud PC Frontline 2vCPU/8GB/128GB	a0249b00-1234-1234-1234	W365-DEV Development Cloud PC FL	
PRD-test1-01387	FrontlineDedicated	TRUE		2025-06-05 07:45		test1@cognitionit.com	test1@cognitionit.com	057efbfe-1234	Cloud PC Frontline 2vCPU/8GB/128GB	db320d66-1234-1234-1234	W365-PRD Production Cloud PC	
PRD-test2-0EU22	FrontlineDedicated	FALSE	2025-11-07 12:57	11/14/2024 10:04:12 AM	35	test2@cognitionit.com	test2@cognitionit.com	057efbfe-1234	Cloud PC Frontline 2vCPU/8GB/128GB	db320d66-1234-1234-1234	W365-PRD Production Cloud PC	

W365 CloudPC License Management (Postman)



HTTP 00. GraphAPI Demos / Windo... / 03. License Management / CloudPCReports: getTotalAggregatedRemoteConnectionReports Save Share

POST

https://graph.microsoft.com/beta/deviceManagement/virtualEndpoint/reports/getTotalAggregatedRemoteConnectionReports/?\$count=1...

Send

Params

Authorization

Headers (12)

Body

Scripts

Settings

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

☒ JSON

Cookies

Beautify

```
1 {
2   "skip": 0,
3   "select": [],
4   "top": 1,
5   "search": "",
6   "orderBy": [
7     "TotalUsageInHour"
8   ]
9 }
```

ManagedDeviceName	DeviceType	NeverSignedIn	LastActiveTime	CreatedDate	DaysSinceLastSignIn	UserPrincipalName	contactEmail	ServicePlanId	ServicePlanName	provisioningPolicyGroupId	provisioningPolicyGroupName	ExpectedEnterpriseLicenseGroupName
TST-test0-B5JM4	Enterprise	TRUE		2025-12-05 12:07		test0@cognitionittst.onmicro	test0@cognitionit.com	2de9c682-1234-1234-1234-1234	Cloud PC Enterprise 4vCPU/16GB/128GB	736b9fc8-1234-1234-1234-1234	W365-TST Test Cloud PC	W365-LIC-4vCPU-16GB-128GB
TST-test1-KP47C	Enterprise	FALSE	6/20/2025 2:37:02 PM	2025-12-05 12:07	56	test1@cognitionittst.onmicro	test1@cognitionit.com	69dc175c-1234-1234-1234-1234	Cloud PC Enterprise 8vCPU/32GB/256GB	736b9fc8-1234-1234-1234-1234	W365-TST Test Cloud PC	W365-LIC-8vCPU-32GB-256GB
DEV-test0-VEOL3	Enterprise	FALSE	7/16/2025 9:24:11 AM	7/14/2023 9:37:07 AM	30	test0@cognitionitdev.onmicrc	test0@cognitionit.com	2de9c682-1234-1234-1234-1234	Cloud PC Enterprise 4vCPU/16GB/128GB	b8483202-1234-1234-1234-1234	W365-DEV Development Cloud PC	W365-LIC-4vCPU-16GB-128GB
DEV-test1-ZSSX8	FrontlineDedicated	FALSE	7/16/2025 5:55:31 PM	2025-09-07 08:38	30	test1@cognitionitdev.onmicrc	test1@cognitionit.com	057efbfe-1234-1234-1234-1234	Cloud PC Frontline 2vCPU/8GB/128GB	a0249b00-1234-1234-1234-1234	W365-DEV Development Cloud PC FL	
DEV-test2-FAUF8	Enterprise	FALSE	6/30/2025 7:09:21 AM	2024-08-04 06:28	46	test2@cognitionitdev.onmicrc	test2@cognitionit.com	3efff3fe-1234-1234-1234-1234	Cloud PC Enterprise 2vCPU/8GB/128GB	b8483202-1234-1234-1234-1234	W365-DEV Development Cloud PC	W365-LIC-2vCPU-8GB-128GB
DEV-test3-D1U44	FrontlineDedicated	TRUE		8/13/2025 8:08:52 AM		test3@cognitionitdev.onmicrc	test3@cognitionit.com	057efbfe-1234-1234-1234-1234	Cloud PC Frontline 2vCPU/8GB/128GB	a0249b00-1234-1234-1234-1234	W365-DEV Development Cloud PC FL	
PRD-test1-01387	FrontlineDedicated	TRUE		2025-06-05 07:45		test1@cognitionit.com	test1@cognitionit.com	057efbfe-1234-1234-1234-1234	Cloud PC Frontline 2vCPU/8GB/128GB	db320d66-1234-1234-1234-1234	W365-PRD Production Cloud PC	
PRD-test2-0EU22	FrontlineDedicated	FALSE	2025-11-07 12:57	11/14/2024 10:04:12 AM	35	test2@cognitionit.com	test2@cognitionit.com	057efbfe-1234-1234-1234-1234	Cloud PC Frontline 2vCPU/8GB/128GB	db320d66-1234-1234-1234-1234	W365-PRD Production Cloud PC	

W365 License Management

PowerShell script challenges



W365 CloudPC License Management

Get-W365CPC-InactiveFor60Days-Report-Data

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

W365 CloudPC License Management

Get-W365CPC-InactiveFor60Days-Report-Data

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

Challenges:

- response data pagination
- response formatting (PSObject)

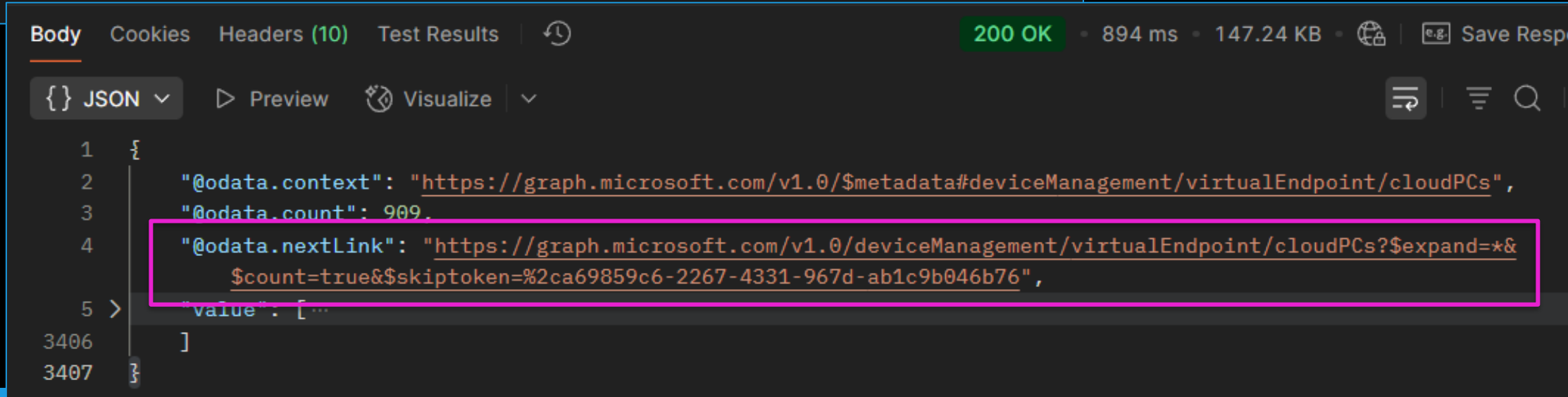
Graph API PowerShell challenge (GET pagination)

Paging Microsoft Graph data in your app

04/03/2025

Paging involves requesting or receiving data in batches. It's a performance technique that's crucial for efficiently handling large datasets and that helps improve the performance of your app and the response time of Microsoft Graph.

Some GET queries against Microsoft Graph return multiple pages of data either due to server-side paging or client-side paging. In this article, we explore how paging works for Microsoft Graph and how you can use it to optimize your applications.

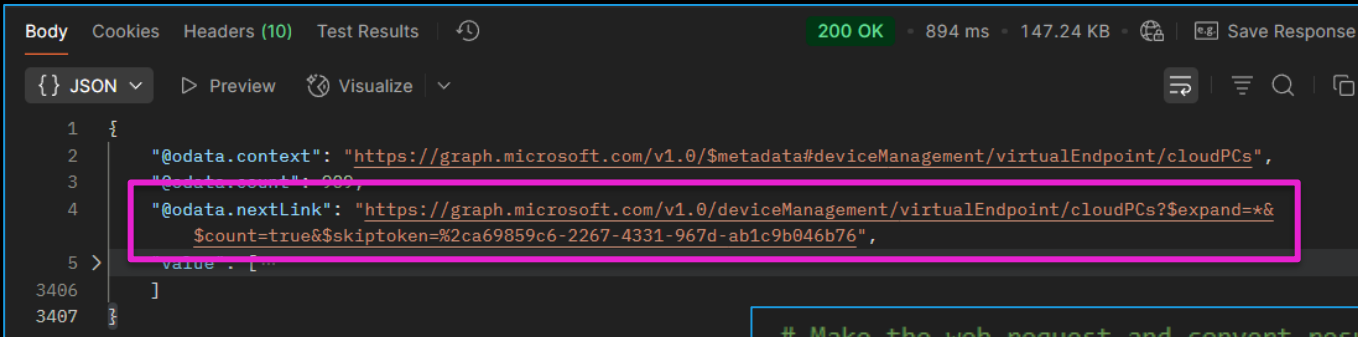


The screenshot shows a REST client interface with the following details:

- Body** tab selected, showing JSON data.
- Status: **200 OK**, 894 ms, 147.24 KB.
- JSON data structure:

```
1 {
2   "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#deviceManagement/virtualEndpoint/cloudPCs",
3   "@odata.count": 909,
4   "@odata.nextLink": "https://graph.microsoft.com/v1.0/deviceManagement/virtualEndpoint/cloudPCs?$expand=*&$count=true&$skiptoken=%2ca69859c6-2267-4331-967d-ab1c9b046b76",
5   "value": [ ...
3406 ]
3407 }
```
- The `@odata.nextLink` value is highlighted with a pink box.

Graph API PowerShell challenge (GET pagination)



```
Body Cookies Headers (10) Test Results 200 OK 894 ms 147.24 KB Save Response
JSON Preview Visualize
1 {
2   "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#deviceManagement/virtualEndpoint/cloudPCs",
3   "@odata.count": 999,
4   "@odata.nextLink": "https://graph.microsoft.com/v1.0/deviceManagement/virtualEndpoint/cloudPCs?$expand=*&$count=true&$skiptoken=%2ca69859c6-2267-4331-967d-ab1c9b046b76",
5   "value": [
3406   ]
3407 }
```

```
# Make the web request and convert result content to PSObject to retrieve the Windows365 CloudPCs
$webrequestResult = ConvertFrom-Json -InputObject $($($Invoke-WebRequest @webRequestParams).Content)

# Create an array to store all CloudPCs
$allCloudPCs = @()
$allCloudPCs += $webrequestResult.value

# Resolve Graph API pagination to retrieve all CloudPCs
while ($null -ne $webrequestResult.'@odata.nextLink') {
    $uri = $webrequestResult.'@odata.nextLink'
    $webRequestParams = @{
        Uri      = $uri
        Headers  = $authHeader
        Method   = $method
    }
    $webrequestResult = ConvertFrom-Json -InputObject $($($Invoke-WebRequest @webRequestParams).Content)
    # Store next page of CloudPCs in the allCloudPCs array
    $allCloudPCs += $($webrequestResult.value)
}

# Return the array of all CloudPCs
return $allCloudPCs
```

Graph API PowerShell challenge (GET pagination)

```
Body Cookies Headers (10) Test Results 200 OK 894 ms 147.24 KB Save Response
JSON Preview Visualize
1 {
2   "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#deviceManagement/virtualEndpoint/cloudPCs",
3   "@odata.count": 909,
4   "@odata.nextLink": "https://graph.microsoft.com/v1.0/deviceManagement/virtualEndpoint/cloudPCs?$expand=*&$count=true&$skiptoken=%2ca69859c6-2267-4331-967d-ab1c9b046b76",
5   "value": [ ...
3406 ]
3407 }
```

```
# Make the web request and convert result content to PSObject to retrieve the Windows365 CloudPCs
$webrequestResult = ConvertFrom-Json -InputObject $($($Invoke-WebRequest @webRequestParams).Content)

# Create an array to store all CloudPCs
$allCloudPCs = @()
$allCloudPCs += $webrequestResult.value

# Resolve Graph API pagination to retrieve all CloudPCs
while ($null -ne $webrequestResult.'@odata.nextLink') {
    $uri = $webrequestResult.'@odata.nextLink'
    $webRequestParams = @{
        Uri = $uri
        Headers = $authHeader
        Method = $method
    }
    $webrequestResult = ConvertFrom-Json -InputObject $($($Invoke-WebRequest @webRequestParams).Content)
    # Store next page of CloudPCs in the allCloudPCs array
    $allCloudPCs += $($webrequestResult.value)
}

# Return the array of all CloudPCs
return $allCloudPCs
```

Graph API PowerShell challenge (POST pagination)

```
Body Cookies Headers (7) Test Results
{ } JSON Visualize
1 {
2   "TotalRowCount": 906,
3   "Schema": [ ...
68 ],
69 "Values": [ ...
1870 ]
1871 }
```



NO @odata.nextLink !!!!!

Graph API PowerShell challenge (POST pagination)



Paging data from the Microsoft Graph API using the `POST` method—especially for endpoints like **Cloud PC reports**—requires a slightly different approach than traditional `GET` requests. Here's how you can handle pagination using the `skip` parameter in this context:

Scenario: Paging Cloud PC Reports via POST

Microsoft Graph's reporting endpoints (like `/reports/cloudPCPerformance`, `/reports/cloudPCConnection`, etc.) often use `POST` for complex queries. These endpoints typically support pagination via the `skip` parameter in the request body.

3. Loop Through Pages

To get all data, loop through the pages by increasing `skip` :

Python ^

 Copy

```
skip = 0
top = 100
results = []

while True:
    response = post_to_graph_api(skip=skip, top=top)
    data = response["value"]
    results.extend(data)

    if len(data) < top:
        break # No more data
    skip += top
```

Body Cookies Headers (7) Test Results ↺

{ } JSON ▾

 Visualize ▾

```
1 {
2   "TotalRowCount": 906,
3   "Schema": [ ...
68 ],
69 "Values": [ ...
1870 ]
1871 }
```



NO @odata.nextLink !!!!!

Graph API PowerShell challenge (POST pagination)

```
Body Cookies Headers (7) Test Results
JSON Visualize
1 {
2   "TotalRowCount": 906,
3   "Schema": [ ...
68 ],
69 "Values": [ ...
1870 ]
1871 }
```



```
# Create an array to store all AggregatedRemoteConnections
[array]$rawAggregatedRemoteConnections = @()

do {
    # Make the web request to retrieve the Total Aggregated Remote Connection Reports
    $webRequestParams = @{
        Uri      = $uri
        Headers  = $authHeader
        Method   = $method
        Body     = $(ConvertTo-Json -InputObject $bodyObject -Depth 100)
    }

    # Use Invoke-RestMethod to handle base64-encoded webrequest.content
    $webrequestResults = Invoke-RestMethod @webRequestParams

    # Process the result of the web request
    [int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
    foreach ($item in $webrequestResults.Values) {
        $aggregatedRemoteConnection = @{}
        for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
            $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])
        }
        $rawAggregatedRemoteConnections += $aggregatedRemoteConnection
    }

    # Update skip and check for pagination
    $retrievedRows = ($webrequestResults.Values | Measure-Object).Count
    $skip += $retrievedRows
    $totalRowCount = $webrequestResults.TotalRowCount

    # Update the request body for the next page
    $bodyObject["skip"] = $skip
} while ($skip -lt $totalRowCount)

# Return the array of all Aggregated Remote Connections
return $rawAggregatedRemoteConnections
```

Graph API PowerShell challenge (POST pagination)

```
Body Cookies Headers (7) Test Results
JSON Visualize
1 {
2   "TotalRowCount": 906,
3   "Schema": [ ...
68 ],
69 "Values": [ ...
1870 ]
1871 }
```



```
# Create an array to store all AggregatedRemoteConnections
[array]$rawAggregatedRemoteConnections = @()

do {
    # Make the web request to retrieve the Total Aggregated Remote Connection Reports
    $webRequestParams = @{
        Uri      = $uri
        Headers  = $authHeader
        Method   = $method
        Body     = $(ConvertTo-Json -InputObject $bodyObject -Depth 100)
    }

    # Use Invoke-RestMethod to handle base64-encoded webrequest.content
    $webrequestResults = Invoke-RestMethod @webRequestParams

    # Process the result of the web request
    [int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
    foreach ($item in $webrequestResults.Values) {
        $aggregatedRemoteConnection = @{}
        for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
            $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])
        }
        $rawAggregatedRemoteConnections += $aggregatedRemoteConnection
    }

    # Update skip and check for pagination
    $retrievedRows = ($webrequestResults.Values | Measure-Object).Count
    $skip += $retrievedRows
    $totalRowCount = $webrequestResults.TotalRowCount

    # Update the request body for the next page
    $bodyObject["skip"] = $skip
    while ($skip -lt $totalRowCount)

    # Return the array of all Aggregated Remote Connections
    return $rawAggregatedRemoteConnections
}
```

Graph API PowerShell challenge (response -> PSObject)

```
Body Cookies Headers (10) Test Results 200 OK • 882 ms • 1.54 KB Save Response
{} JSON Preview Visualize
1 {
2   "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#deviceManagement/virtualEndpoint/cloudPCs",
3   "@odata.count": 2,
4   "@odata.nextLink": "https://graph.microsoft.com/v1.0/deviceManagement/virtualEndpoint/cloudPCs?$expand=*&$count=true&
5     $filter=startswith(userPrincipalName%2c+%27test01%27)&$top=1&$skiptoken=%2c1c60d512-df61-4e6a-9e11-3088259bf376",
6   "value": [
7     {
8       "id": "c15e0192-1977-4514-a7e4-3095225d4452",
9       "displayName": "CPC-W365-Frontline-Provisioning - test01",
10      "imageDisplayName": "Windows 11 Enterprise + Microsoft 365 Apps 24H2",
11      "provisioningPolicyId": "a5a00319-eb6b-478a-bcda-9d76bcd7fdf5",
12      "provisioningPolicyName": "CPC-W365-Frontline-Provisioning",
13      "onPremisesConnectionName": "",
14      "servicePlanId": "50ef7026-6174-40ba-bff7-f0e4fcddb65",
15      "servicePlanName": "Cloud PC Frontline 2vCPU/8GB/256GB",
16      "userPrincipalName": "test01@cognitionitdev.onmicrosoft.com",
17      "lastModifiedDate": "2025-08-31T09:55:50Z",
18      "managedDeviceId": null,
19      "managedDeviceName": null,
20      "aadDeviceId": null,
21      "gracePeriodEndDateTime": null,
22      "provisioningType": "shared"
23    }
24  ]
25 }
```

Graph API PowerShell challenge (response -> PSObject)

```
# Make the web request and convert result content to PSObject to retrieve the Windows365 CloudPCs
$webrequestResult = ConvertFrom-Json -InputObject $($($Invoke-WebRequest @webRequestParams).Content)

# Create an array to store all CloudPCs
$allCloudPCs = @()
$allCloudPCs += $webrequestResult.value

# Resolve Graph API pagination to retrieve all CloudPCs
while ($null -ne $webrequestResult.'@odata.nextLink') {
    $uri = $webrequestResult.'@odata.nextLink'
    $webRequestParams = @{ ...
}
$webrequestResult = ConvertFrom-Json -InputObject $($($Invoke-WebRequest @webRequestParams).Content)
# Store next page of CloudPCs in the allCloudPCs array
$allCloudPCs += $($webrequestResult.value)
}

# Return the array of all CloudPCs
return $allCloudPCs
```

Graph API PowerShell challenge (response -> PSObject)

```
W365iicenseReport.ps1 X
C: > Temp > W365iicenseReport.ps1 > ...
591 $w365CloudPCs = Get-W365CloudPCs -AccessToken $($graphContext.access_token)
592
```

```
PROBLEMS  TERMINAL  PORTS  GITLENS  AZURE  OUTPUT  DEBUG CONSOLE

● PS C:\Temp> $w365CloudPCs[0]

id                : c15e0192-1977-4514-a7e4-3095225d4452
displayName        : CPC-W365-Frontline-Provisioning - test01
imageDisplayName   : Windows 11 Enterprise + Microsoft 365 Apps 24H2
provisioningPolicyId : a5a00319-eb6b-478a-bcda-9d76bcd7fdf5
provisioningPolicyName : CPC-W365-Frontline-Provisioning
onPremisesConnectionName :
servicePlanId      : 50ef7026-6174-40ba-bff7-f0e4fcddb65
servicePlanName     : Cloud PC Frontline 2vCPU/8GB/256GB
status              : provisioning
userPrincipalName   : test01@cognitionitdev.onmicrosoft.com
lastModifiedDateTime : 2025-08-31 09:53:38
managedDeviceId     :
managedDeviceName   :
aadDeviceId         :
gracePeriodEndDateTime :
servicePlanType      : enterprise
diskEncryptionState  : notAvailable
provisioningType      : sharedByUser
statusDetails        :
statusDetail         :
partnerAgentInstallResults : {}

PS C:\Temp>
```

Graph API PowerShell challenge (response -> PSObject)

```
Body Cookies Headers (7) Test Results
JSON Visualize
1 {
2   "TotalRowCount": 2,
3   "Schema": [
4     {
5       "Column": "CloudPcId",
6       "PropertyType": "String"
7     },
8     {
9       "Column": "ManagedDeviceName",
10      "PropertyType": "String"
11     },
12     {
13       "Column": "UserPrincipalName",
14       "PropertyType": "String"
15     },
16     {
17       "Column": "DaysSinceLastSignIn",
18       "PropertyType": "Int64"
19     },
20     {
21       "Column": "TotalUsageInHour",
22       "PropertyType": "Double"
23     },
24   ]
25 }
```

```
Body Cookies Headers (7) Test Results
JSON Visualize
1 {
2   "TotalRowCount": 2,
3   "Schema": [ ...
68 ],
69 "Values": [
70   [ ...
87 ],
88 [
89   "627a21fe-a554-47ca-9a3c-f14fe8c3e99b",
90   "CPC-test0-N61MG",
91   "test01@cognitionitdev.onmicrosoft.com",
92   null,
93   0.0,
94   "645f6084-98ba-4ed4-9299-817ed164860d",
95   null,
96   true,
97   "50ef7026-6174-40ba-bff7-f0e4fcddbf65",
98   "Cloud PC Frontline 2vCPU/8GB/256GB",
99   "Cloud PC Frontline 2vCPU/8GB/256GB",
100  "2025-08-27T20:37:09",
101  "northeurope",
102  "FrontlineDedicated",
103  "CPC-W365-Frontline-Provisioning",
104  "a5a00319-eb6b-478a-bcda-9d76bcd7fdf5"
105 ],
106 ]
}
```

Graph API PowerShell challenge (response -> PSObject)

```
# Create an array to store all AggregatedRemoteConnections
[array]$rawAggregatedRemoteConnections = @()

do {
    # Make the web request to retrieve the Total Aggregated Remote Connection Reports
    $webRequestParams = @{ ...
    }

    # Use Invoke-RestMethod to handle base64-encoded webrequest.content
    $webrequestResults = Invoke-RestMethod @webRequestParams

    # Process the result of the web request
    [int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
    foreach ($item in $webrequestResults.Values) {
        $aggregatedRemoteConnection = @{}
        for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
            $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])
        }
        $rawAggregatedRemoteConnections += $aggregatedRemoteConnection
    }
} while ($skip -lt $totalRowCount)

# Return the array of all Aggregated Remote Connections
return $rawAggregatedRemoteConnections
```


Graph API PowerShell challenge (response -> PSObject)

```
# Create an array to store all AggregatedRemoteConnections
[array]$rawAggregatedRemoteConnections = @()

do {
    # Make the web request to retrieve the Total Aggregated Remote Connection Reports
    $webRequestParams = @{ ...
    }

    # Use Invoke-RestMethod to handle base64-encoded webrequest.content
    $webrequestResults = Invoke-RestMethod @webRequestParams

    # Process the result of the web request
    [int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
    foreach ($item in $webrequestResults.Values) {
        $aggregatedRemoteConnection = @{}
        for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
            $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])
        }
        $rawAggregatedRemoteConnections += $aggregatedRemoteConnection
    }
} while ($skip -lt $totalRowCount)

# Return the array of all Aggregated Remote Connections
return $rawAggregatedRemoteConnections
```

```
"Schema": [
  {
    "Column": "CloudPcId",
    "PropertyType": "String"
  },
  {
    "Column": "ManagedDeviceName",
    "PropertyType": "String"
  }
],
```

```
"Values": [
  [ ...
  ],
  [
    "627a21fe-a554-47ca-9a3c-f14fe8c3e99b",
    "CPC-test0-N61MG",
  ]
]
```


Graph API PowerShell challenge (response -> PSObject)

C: > Temp > W365iicenseReport.ps1 > ...

```
607 # Get Intune report data for Total Aggregated Remote Connections
608 $rawAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReport -AccessToken $($graphContext.access_token)
609 $enrichedAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReportWithEnrichedData -AccessToken $($graphContext.access_token) `
610 -ServicePlanSearchtable $w365ServicePlanSearchtable -LicenseGroupSearchtable $w365LicenseGroupsSearchTable `
611 -CloudPcSearchtable $w365CloudPCs -ProvisioningPolicySearchtable $w365ProvisioningPoliciesSearchTable
612
```

PROBLEMS TERMINAL PORTS GITLENS AZURE OUTPUT DEBUG CONSOLE

PS C:\Temp> \$rawAggregatedRemoteConnections[1] | ConvertTo-Json -Depth 10 | ConvertFrom-Json

```
LastActiveTime      :
CreatedDate         : 2025-08-27 20:37:09
ManagedDeviceName  : CPC-test0-N61MG
ServicePlanId       : 50ef7026-6174-40ba-bff7-f0e4fcddb65
UserPrincipalName   : test01@cognitionitdev.onmicrosoft.com
CloudPcId           : 627a21fe-a554-47ca-9a3c-f14fe8c3e99b
DeviceType          : FrontlineDedicated
CloudPCSize         : Cloud PC Frontline 2vCPU/8GB/256GB
IntuneDeviceId      : 645f6084-98ba-4ed4-9299-817ed164860d
PcType              : Cloud PC Frontline 2vCPU/8GB/256GB
DaysSinceLastSignIn :
PolicyName           : CPC-W365-Frontline-Provisioning
NeverSignedIn       : True
HostRegion           : northeurope
TotalUsageInHour    : 0
```

PS C:\Temp>

W365 CloudPC License Management

Get-W365CPC-InactiveFor60Days-Report-Data

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

W365 CloudPC License Management

Get-W365CPC-InactiveFor60Days-Report-Data

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

Challenges:

- add data to PSObject
- optimize additional API calls

Graph API PowerShell challenge (enrich PSObject)

```
# Create an array to store all AggregatedRemoteConnections
[array]$rawAggregatedRemoteConnections = @()

do {
    # Make the web request to retrieve the Total Aggregated Remote Connection Reports
    $webRequestParams = @{ ...
    }

    # Use Invoke-RestMethod to handle base64-encoded webrequest.content
    $webrequestResults = Invoke-RestMethod @webRequestParams

    # Process the result of the web request
    [int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
    foreach ($item in $webrequestResults.Values) {
        $aggregatedRemoteConnection = @{}
        for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
            $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])
        }
        $rawAggregatedRemoteConnections += $aggregatedRemoteConnection
    }
} while ($skip -lt $totalRowCount)

# Return the array of all Aggregated Remote Connections
return $rawAggregatedRemoteConnections
```

Graph API PowerShell challenge (enrich PSObject)

```
# Process the result of the web request
[int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
foreach ($item in $webrequestResults.Values) {
    $aggregatedRemoteConnection = @{}
    for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
        $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])

        # Add ServicePlan details
        if ($webrequestResults.Schema[$i].Column -eq 'ServicePlanId') {
            $servicePlan = $ServicePlanSearchTable.where({ $_.id -eq $item[$i] })
            $aggregatedRemoteConnection.Add('ServicePlanName', $servicePlan.displayName)
            $aggregatedRemoteConnection.Add('ServicePlanType', $servicePlan.type)
            # Add ServicePlanProvisioningType and ExpectedEnterpriseLicenseGroupName
            switch ($servicePlan.provisioningType) {
                'shared' {
                    $ServicePlanProvisioningType = 'Frontline'
                    $EnterpriseLicenseGroupName = ''
                }
                'dedicated' {
                    $ServicePlanProvisioningType = 'Enterprise'
                    $EnterpriseLicenseGroupName = "W365-LIC-$(($servicePlan.vCpuCount)vCPU-")
                    $EnterpriseLicenseGroupName += " $(($servicePlan.ramInGB)GB-")
                    $EnterpriseLicenseGroupName += " $(($servicePlan.storageInGB)GB)"
                }
                default {
                    $ServicePlanProvisioningType = 'Unknown'
                    $EnterpriseLicenseGroupName = ''
                }
            }
            $aggregatedRemoteConnection.Add('ServicePlanProvisioningType', $ServicePlanProvisioningType)
            $aggregatedRemoteConnection.Add('ExpectedEnterpriseLicenseGroupName', $EnterpriseLicenseGroupName)
            # search EnterpriseLicenseGroupId in $EnterpriseLicenseGroupsSearchTable
            $licenseGroup = $licenseGroupSearchTable.where({ $_.displayName -eq $EnterpriseLicenseGroupName })
            $aggregatedRemoteConnection.Add('ExpectedEnterpriseLicenseGroupId', $($licenseGroup.id))
        }

        # Add CloudPC details
        if ($webrequestResults.Schema[$i].Column -eq 'CloudPcId') {
            $cloudPC = $CloudPcSearchTable.where({ $_.id -eq $item[$i] })
            $aggregatedRemoteConnection.Add('CloudPCName', $cloudPC.displayName)
            $aggregatedRemoteConnection.Add('ProvisioningPolicyId', $cloudPC.provisioningPolicyId)
            $aggregatedRemoteConnection.Add('ProvisioningPolicyName', $cloudPC.provisioningPolicyName)
            $aggregatedRemoteConnection.Add('ProvisioningType', $cloudPC.provisioningType)
            $aggregatedRemoteConnection.Add('lastModifiedDateTime', $cloudPC.lastModifiedDateTime)
            $aggregatedRemoteConnection.Add('provisioningStatus', $cloudPC.status)
            # Add provisioning policy groupname & groupid to dataset
            $assignmentServicePlanId = $cloudPC.servicePlanId
            $assignmentProvisioningPolicyId = $cloudPC.provisioningPolicyId
            $provisioningPolicy = $ProvisioningPolicySearchTable.where({ $_.id -eq $assignmentProvisioningPolicyId }
            -and (($_.assignmentServicePlanId -eq $assignmentServicePlanId) -or ($null -eq $_.assignmentServicePlanId) ))
            $aggregatedRemoteConnection.Add('provisioningPolicyGroupId', $provisioningPolicy.assignmentGroupId)
            $aggregatedRemoteConnection.Add('provisioningPolicyGroupName', $provisioningPolicy.assignmentGroupName)
            $aggregatedRemoteConnection.Add('aadDeviceId', $cloudPC.aadDeviceId)
            # Convert UPN to contact email address
            $userPrincipalName = $cloudPC.userPrincipalName
            if ($userPrincipalName.EndsWith('cognitionitdev.onmicrosoft.com')) {
                $userPrincipalName = $userPrincipalName.Replace('cognitionitdev.onmicrosoft.com', 'cognitionit.com')
            }
            if ($userPrincipalName.EndsWith('cognitionittst.onmicrosoft.com')) {
                $userPrincipalName = $userPrincipalName.Replace('cognitionittst.onmicrosoft.com', 'cognitionit.com')
            }
            $contactEmail = $userPrincipalName
            $aggregatedRemoteConnection.Add('contactEmail', $contactEmail)
        }
    }
    $allAggregatedRemoteConnections += $aggregatedRemoteConnection
}
```

Graph API PowerShell challenge (enrich PSObject)

```
# Process the result of the web request
[int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
foreach ($item in $webrequestResults.Values) {
    $aggregatedRemoteConnection = @{}
    for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
        $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])

        # Add ServicePlan details
        if ($webrequestResults.Schema[$i].Column -eq 'ServicePlanId') {
            $servicePlan = $ServicePlanSearchtable.where({ $_.id -eq $item[$i] })
            $aggregatedRemoteConnection.Add('ServicePlanName', $servicePlan.displayName)
            $aggregatedRemoteConnection.Add('ServicePlanType', $servicePlan.type)
        }
    }
}
```

Graph API PowerShell challenge (enrich PSObject)

```
# Process the result of the web request
[int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
foreach ($item in $webrequestResults.Values) {
    $aggregatedRemoteConnection = @{}
    for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
        $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])

        # Add ServicePlan details
        if ($webrequestResults.Schema[$i].Column -eq 'ServicePlanId') {
            $servicePlan = $ServicePlanSearchtable.where({ $_.id -eq $item[$i] })
            $aggregatedRemoteConnection.Add('ServicePlanName', $servicePlan.displayName)
            $aggregatedRemoteConnection.Add('ServicePlanType', $servicePlan.type)
        }
    }
}
```

Graph API PowerShell challenge (enrich PSObject)

```
# Process the result of the web request
[int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
foreach ($item in $webrequestResults.Values) {
    $aggregatedRemoteConnection = @{}
    for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
        $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])

        # Add ServicePlan details
        if ($webrequestResults.Schema[$i].Column -eq 'ServicePlanId') {
            $servicePlan = $ServicePlanSearchtable.where({ $_.id -eq $item[$i] })
            $aggregatedRemoteConnection.Add('ServicePlanName', $servicePlan.displayName)
            $aggregatedRemoteConnection.Add('ServicePlanType', $servicePlan.type)
        }
    }
}
```


Graph API PowerShell challenge (enrich PSObject)

```
# Process the result of the web request
[int]$totalColumns = ($webrequestResults.Schema | Measure-Object).Count
foreach ($item in $webrequestResults.Values) {
    $aggregatedRemoteConnection = @{}
    for ($i = 0; $i -lt ($totalColumns - 1); $i++) {
        $aggregatedRemoteConnection.Add($webrequestResults.Schema[$i].Column, $item[$i])

        # Add ServicePlan details
        if ($webrequestResults.Schema[$i].Column -eq 'ServicePlanId') {
            $servicePlan = $ServicePlanSearchtable.where({ $_.id -eq $item[$i] })
            $aggregatedRemoteConnection.Add('ServicePlanName', $servicePlan.displayName)
            $aggregatedRemoteConnection.Add('ServicePlanType', $servicePlan.type)
        }
    }
}
```



```
CloudPcId      : 627a21fe-a554-47ca-9a3c-f14fe8c3e99b
ManagedDeviceName : CPC-test0-N61MG
UserPrincipalName : test01@cognitionitdev.onmicrosoft.com
ServicePlanId    : 50ef7026-6174-40ba-bff7-f0e4fcddb65
ServicePlanName   : Cloud PC Frontline 2vCPU/8GB/256GB
ServicePlanType   : enterprise
```

Graph API PowerShell challenge (enrich PSObject)

```
# Add ServicePlanProvisioningType and ExpectedEnterpriseLicenseGroupName
switch ($servicePlan.provisioningType) {
    'shared' {
        $ServicePlanProvisioningType = 'Frontline'
        $EnterpriseLicenseGroupName = ''
    }
    'dedicated' {
        $ServicePlanProvisioningType = 'Enterprise'
        $EnterpriseLicenseGroupName = "W365-LIC-$(($servicePlan.vCpuCount)vCPU-"
        $EnterpriseLicenseGroupName += "$(($servicePlan.ramInGB)GB-"
        $EnterpriseLicenseGroupName += "$(($servicePlan.storageInGB)GB"
    }
    default {
        $ServicePlanProvisioningType = 'Unknown'
        $EnterpriseLicenseGroupName = ''
    }
}

$aggregatedRemoteConnection.Add('ServicePlanProvisioningType', $ServicePlanProvisioningType)
$aggregatedRemoteConnection.Add('ExpectedEnterpriseLicenseGroupName', $EnterpriseLicenseGroupName)
# search EnterpriseLicenseGroupId in $w365LicenseGroupsSearchTable
$licenseGroup = $LicenseGroupSearchtable.where({ $_.displayName -eq $EnterpriseLicenseGroupName })
$aggregatedRemoteConnection.Add('ExpectedEnterpriseLicenseGroupId', $($licenseGroup.id))
}
```



CloudPcId	: 627a21fe-a554-47ca-9a3c-f14fe8c3e99b
ManagedDeviceName	: CPC-test0-N61MG
UserPrincipalName	: test01@cognitionitdev.onmicrosoft.com
ServicePlanProvisioningType	: Enterprise
ExpectedEnterpriseLicenseGroupName	: W365-LIC-2vCPU-8GB-256GB
ExpectedEnterpriseLicenseGroupId	: 12345678-1234-1234-1234-123456789012

Graph API PowerShell challenge (enrich PSObject)

```
# Add CloudPC details
if ($webrequestResults.Schema[$i].Column -eq 'CloudPcId') {
    $cloudPC = $CloudPcSearchtable.where({ $_.id -eq $item[$i] })
    $aggregatedRemoteConnection.Add('CloudPCName', $cloudPC.displayName)
    $aggregatedRemoteConnection.Add('ProvisioningPolicyId', $cloudPC.provisioningPolicyId)
    $aggregatedRemoteConnection.Add('ProvisioningPolicyName', $cloudPC.provisioningPolicyName)
    $aggregatedRemoteConnection.Add('provisioningType', $cloudPC.provisioningType)
    $aggregatedRemoteConnection.Add('lastModifiedDateTime', $cloudPC.lastModifiedDateTime)
    $aggregatedRemoteConnection.Add('provisioningStatus', $cloudPC.status)
    # Add provisioning policy groupname & groupid to dataset
    $assignmentServicePlanId = $cloudPC.servicePlanId
    $assignmentProvisioningPolicyId = $cloudPC.provisioningPolicyId
    $provisioningPolicy = $ProvisioningPolicySearchtable.where({ ($_.id -eq $assignmentProvisioningPolicyId) `
        -and (($_.assignmentServicePlanId -eq $assignmentServicePlanId) -or ($null -eq $_.assignmentServicePlanId) ) })
    $aggregatedRemoteConnection.Add('provisioningPolicyGroupId', $provisioningPolicy.assignmentGroupId)
    $aggregatedRemoteConnection.Add('provisioningPolicyGroupName', $provisioningPolicy.assignmentGroupName)
    $aggregatedRemoteConnection.Add('aadDeviceId', $cloudPC.aadDeviceId)
    # Convert UPN to contact email address
    $userPrincipalName = $cloudPC.userPrincipalName
    if ($userPrincipalName.EndsWith('cognitionitdev.onmicrosoft.com')) {
        $userPrincipalName = $userPrincipalName.Replace('cognitionitdev.onmicrosoft.com', 'cognitionit.com')
    }
    if ($userPrincipalName.EndsWith('cognitionittst.onmicrosoft.com')) {
        $userPrincipalName = $userPrincipalName.Replace('cognitionittst.onmicrosoft.com', 'cognitionit.com')
    }
    $contactEmail = $userPrincipalName
    $aggregatedRemoteConnection.Add('contactEmail', $contactEmail)
}
```



CloudPcId	: 627a21fe-a554-47ca-9a3c-f14fe8c3e99b
CloudPCName	: CPC-W365-Frontline-Provisioning - test01
ProvisioningPolicyId	: a5a00319-eb6b-478a-bcda-9d76bcd7fdf5
ProvisioningPolicyName	: CPC-W365-Frontline-Provisioning
provisioningType	: sharedByUser
lastModifiedDateTime	: 2025-08-27 23:14:00
provisioningStatus	: provisioned
ProvisioningPolicyGroupId	: 12345678-1234-1234-1234-123456789012
provisioningPolicyGroupName	: {\$null, grp-prv-W365-DEV-Users}
aadDeviceId	: 12345678-1234-1234-1234-123456789012
UserPrincipalName	: test01@cognitionitdev.onmicrosoft.com
contactEmail	: test01@cognitionit.com

Graph API PowerShell challenge (searchtables)

```
C: > Temp > W365iicenseReport.ps1 > function Get-W365AggregatedRemoteConnectionReportWithEnrichedData ()  
394 function Get-W365AggregatedRemoteConnectionReportWithEnrichedData {  
409     #region Create URI for Graph call to collect the Total Aggregated Remote Connection Reports  
455  
456     # Add ServicePlan details  
457     if ($webrequestResults.Schema[$i].Column -eq 'ServicePlanId') {  
458         $servicePlan = $ServicePlanSearchtable.where({ $_.id -eq $item[$i] })  
459         $aggregatedRemoteConnection.Add('ServicePlanName', $servicePlan.displayName)  
460         $aggregatedRemoteConnection.Add('ServicePlanType', $servicePlan.type)  
461     }
```

PROBLEMS TERMINAL PORTS GITLENS AZURE OUTPUT DEBUG CONSOLE

```
PS C:\Temp> $w365ServicePlanSearchtable.where({ $_.id -eq '50ef7026-6174-40ba-bff7-f0e4fcddbf65' })
```

```
id                : 50ef7026-6174-40ba-bff7-f0e4fcddbf65  
displayName        : Cloud PC Frontline 2vCPU/8GB/256GB  
type               : enterprise  
vCpuCount          : 2  
ramInGB            : 8  
storageInGB        : 256  
userProfileInGB    : 25  
supportedSolution  : windows365  
provisioningType   : shared
```

```
PS C:\Temp>
```

Graph API PowerShell challenge (searchtables)

```
# Build the Searchtables
$w365ServicePlanSearchtable = Get-W365ServicePlans -AccessToken $($graphContext.access_token)
$w365ProvisioningPolicyGroupSearchtable = Get-W365ProvisioningPolicyGroups -AccessToken $($graphContext.access_token) `
    -ProvisioningPolicyGroupDisplayName "grp-prv-W365"
$w365LicenseGroupsSearchtable = Get-W365LicenseGroups -AccessToken $($graphContext.access_token) `
    -LicenseGroupDisplayName "grp-lic-W365"
$w365ProvisioningPoliciesSearchtable = Get-W365ProvisioningPolicies -AccessToken $($graphContext.access_token) `
    -ProvisioningPolicyGroupSearchTable $w365ProvisioningPolicyGroupSearchtable `
    -ServicePlanSearchTable $w365ServicePlanSearchtable
$w365CloudPCs = Get-W365CloudPCs -AccessToken $($graphContext.access_token)
```

Graph API PowerShell challenge (searchtables)

```
# Build the Searchtables
$w365ServicePlanSearchtable = Get-W365ServicePlans -AccessToken $($graphContext.access_token)
$w365ProvisioningPolicyGroupSearchtable = Get-W365ProvisioningPolicyGroups -AccessToken $($graphContext.access_token) `
    -ProvisioningPolicyGroupDisplayName "grp-prv-W365"
$w365LicenseGroupsSearchtable = Get-W365LicenseGroups -AccessToken $($graphContext.access_token) `
    -LicenseGroupDisplayName "grp-lic-W365"
$w365ProvisioningPoliciesSearchtable = Get-W365ProvisioningPolicies -AccessToken $($graphContext.access_token) `
    -ProvisioningPolicyGroupSearchTable $w365ProvisioningPolicyGroupSearchtable `
    -ServicePlanSearchTable $w365ServicePlanSearchtable
$w365CloudPCs = Get-W365CloudPCs -AccessToken $($graphContext.access_token)
```

PROBLEMS TERMINAL PORTS GITLENS AZURE OUTPUT DEBUG CONSOLE

```
Windows365 Service Plan SearchTable: 52
Windows365 Provisioning Policy Group SearchTable: 1
Windows365 License Groups SearchTable: 1
Windows365 Provisioning Policies SearchTable: 2
Windows365 CloudPCs SearchTable: 3
PS C:\Temp>
```

W365 License Management

PowerShell script



Graph API PowerShell script (functions)

```
#####  
# Script Functions          #  
#####  
#region W365 license management functions  
  
# Graph Authentication functions  
1 reference  
> function Get-GraphAccessToken { ...  
}  
  
# Search Table functions  
1 reference  
> function Get-W365ServicePlans { ...  
}  
  
1 reference  
> function Get-W365ProvisioningPolicyGroups { ...  
}  
  
1 reference  
> function Get-W365LicenseGroups { ...  
}  
  
1 reference  
> function Get-W365ProvisioningPolicies { ...  
}  
  
1 reference  
> function Get-W365CloudPCs { ...  
}  
# Intune Report functions  
1 reference  
> function Get-W365AggregatedRemoteConnectionReport { ...  
}  
  
1 reference  
> function Get-W365AggregatedRemoteConnectionReportWithEnrichedData { ...  
}  
  
# User functions  
1 reference  
> function Get-W365UserId { ...  
}  
#endregion
```


Graph API PowerShell script (workflow)

```
#####  
# Script workflow #  
#####  
# Get the access token  
$graphContext = Get-GraphAccessToken -TenantId $tenantId -AppId $clientId -AppSecret $clientSecret  
# Build the Searchtables  
$w365ServicePlanSearchtable = Get-W365ServicePlans -AccessToken $($graphContext.access_token)  
$w365ProvisioningPolicyGroupSearchtable = Get-W365ProvisioningPolicyGroups -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupDisplayName "grp-prv-W365"  
$w365LicenseGroupsSearchtable = Get-W365LicenseGroups -AccessToken $($graphContext.access_token) -LicenseGroupDisplayName "grp-lic-W365"  
$w365ProvisioningPoliciesSearchtable = Get-W365ProvisioningPolicies -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupSearchTable $w365ProvisioningPolicyGroupSearchtable -ServicePlanSearchTable $w365ServicePlanSearchtable  
$w365CloudPCs = Get-W365CloudPCs -AccessToken $($graphContext.access_token)  
  
#-----#  
# Retrieve Intune Total Aggregated Remote Connection Report for Windows365 #  
#-----#  
# Get Intune report data for Total Aggregated Remote Connections  
$rawAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReport -AccessToken $($graphContext.access_token)  
$enrichedAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReportWithEnrichedData -AccessToken $($graphContext.access_token) -ServicePlanSearchtable $w365ServicePlanSearchtable -LicenseGroupSearchtable $w365LicenseGroupsSearchtable -CloudPCs $w365CloudPCs  
  
# Export Array of Hashtables to JSON file and CSV  
ConvertTo-Json -InputObject $rawAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json" -Force  
$jsonRawData = Get-Content -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json  
$jsonRawData | Export-Csv -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.csv" -NoTypeInformation  
  
# Export Array of Hashtables to JSON file and CSV  
ConvertTo-Json -InputObject $enrichedAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json" -Force  
$jsonEnrichedData = Get-Content -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json  
$jsonEnrichedData | Export-Csv -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.csv" -NoTypeInformation  
  
#-----#  
# FILTERING Enriched Aggregated Remote Connection Report for Windows365 #  
#-----#  
[int]$daysToAdd = -59  
$checkDate = (Get-Date).AddDays($daysToAdd)  
$dateFormatString = "MM/dd/yyyy hh:mm:ss tt"  
#rule 01: No license check for PRD Cloud PCs | Exclude PRD CloudPCs as they will never expire (for now)  
$filteredJsonData = $jsonEnrichedData  
$filteredJsonData = $filteredJsonData.Where({ ($_.provisioningPolicyName -notlike 'PRD*') })  
  
#rule 02: For never logged on provisioned CPCs (NeverLoggedIn = true) check CreatedDate; if older than date limit, email user that license is revoked  
#rule 03: For logged on provisioned CPCs (NeverLoggedIn = false) check LastLoggedInDate; if older than date limit, email user that license is revoked  
# Filter CloudPCs with (NeverLoggedIn eq $true and CreatedDate lt $checkDate) or (NeverLoggedIn eq $false and LastActiveTime le $checkDate)  
$totalFilteredJsonData = $filteredJsonData.Where({ (($_.NeverSignedIn -eq $true) -and ($_.CreatedDate -lt $checkDate) -and ($_.provisioningStatus -eq 'provisioned')) -or (($_.NeverSignedIn -eq $false) -and ($_.LastActiveTime -le $checkDate) -and ($_.provisioningStatus -eq 'provisioned')) })  
  
# Loop through the filtered data and add the userId to the hashtable  
$enrichedTotalFilteredJsonData = $totalFilteredJsonData  
$enrichedTotalFilteredJsonData | ForEach-Object {  
    $userPrincipalName = $_.UserPrincipalName  
    $userId = Get-W365UserId -AccessToken $($graphContext.access_token) -userPrincipalName $userPrincipalName  
    $_ | Add-Member -MemberType NoteProperty -Name "UserDirectoryObjectId" -Value $userId -Force # add the userId to the hashtable  
}  
  
# Export Array of Hashtables to JSON file  
ConvertTo-Json -InputObject $enrichedTotalFilteredJsonData -Depth 100 | Out-File -FilePath "C:\Temp\W365CloudPCsInactiveLast60Days.json" -Force  
$finalJsonData = Get-Content -Path "C:\Temp\W365CloudPCsInactiveLast60Days.json" -Raw | ConvertFrom-Json  
$finalJsonData | Export-Csv -Path "C:\Temp\W365CloudPCsInactiveLast60Days.csv" -NoTypeInformation -Force
```

Graph API PowerShell script

(workflow)

```
#####
# Script workflow #
#####
# Get the access token
$graphContext = Get-GraphAccessToken -TenantId $tenantId -AppId $ClientId -AppSecret $ClientSecret
# Build the Searchtables
$w365ServicePlanSearchtable = Get-W365ServicePlans -AccessToken $($graphContext.access_token)
$w365ProvisioningPolicyGroupSearchtable = Get-W365ProvisioningPolicyGroups -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupSearchtable $w365ProvisioningPolicyGroupSearchtable
$w365LicenseGroupsSearchtable = Get-W365LicenseGroups -AccessToken $($graphContext.access_token) -LicenseGroupDisplayName "grp-lis"
$w365ProvisioningPoliciesSearchtable = Get-W365ProvisioningPolicies -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupSearchtable $w365ProvisioningPolicyGroupSearchtable
$w365CloudPCs = Get-W365CloudPCs -AccessToken $($graphContext.access_token)

#-----#
# Retrieve Intune Total Aggregated Remote Connection Report for Windows365 #
#-----#
# Get Intune report data for Total Aggregated Remote Connections
$rawAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReport -AccessToken $($graphContext.access_token)
$enrichedAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReportWithEnrichedData -AccessToken $($graphContext.access_token)

# Export Array of Hashtables to JSON file and CSV
ConvertTo-Json -InputObject $rawAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json"
$jsonRawData = Get-Content -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json
$jsonRawData | Export-Csv -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.csv" -NoTypeInformation

# Export Array of Hashtables to JSON file and CSV
ConvertTo-Json -InputObject $enrichedAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json"
$jsonEnrichedData = Get-Content -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json
$jsonEnrichedData | Export-Csv -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.csv" -NoTypeInformation
```

Graph API PowerShell script (workflow)

```
#####  
# Script workflow #  
#####  
# Get the access token  
$graphContext = Get-GraphAccessToken -TenantId $tenantId -AppId $ClientId -AppSecret $ClientSecret  
# Build the Searchtables  
$w365ServicePlanSearchtable = Get-W365ServicePlans -AccessToken $($graphContext.access_token)  
$w365ProvisioningPolicyGroupSearchtable = Get-W365ProvisioningPolicyGroups -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupSearchtable  
$w365LicenseGroupsSearchtable = Get-W365LicenseGroups -AccessToken $($graphContext.access_token) -LicenseGroupDisplayName "grp-1" -LicenseGroupSearchtable  
$w365ProvisioningPoliciesSearchtable = Get-W365ProvisioningPolicies -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupSearchtable  
$w365CloudPCs = Get-W365CloudPCs -AccessToken $($graphContext.access_token)  
  
#-----#  
# Retrieve Intune Total Aggregated Remote Connection Report for Windows365 #  
#-----#  
# Get Intune report data for Total Aggregated Remote Connections  
$rawAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReport -AccessToken $($graphContext.access_token)  
$enrichedAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReportWithEnrichedData -AccessToken $($graphContext.access_token)  
  
# Export Array of Hashtables to JSON file and CSV  
ConvertTo-Json -InputObject $rawAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json"  
$jsonRawData = Get-Content -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json  
$jsonRawData | Export-Csv -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.csv" -NoTypeInformation  
  
# Export Array of Hashtables to JSON file and CSV  
ConvertTo-Json -InputObject $enrichedAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json"  
$jsonEnrichedData = Get-Content -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json  
$jsonEnrichedData | Export-Csv -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.csv" -NoTypeInformation
```


Graph API PowerShell script

(workflow)

```
#####
# Script workflow #
#####
# Get the access token
$graphContext = Get-GraphAccessToken -TenantId $tenantId -AppId $ClientId -AppSecret $ClientSecret
# Build the Searchtables
$w365ServicePlanSearchtable = Get-W365ServicePlans -AccessToken $($graphContext.access_token)
$w365ProvisioningPolicyGroupSearchtable = Get-W365ProvisioningPolicyGroups -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupSearchtable $w365ProvisioningPolicyGroupSearchtable
$w365LicenseGroupsSearchtable = Get-W365LicenseGroups -AccessToken $($graphContext.access_token) -LicenseGroupDisplayName "grp-lis"
$w365ProvisioningPoliciesSearchtable = Get-W365ProvisioningPolicies -AccessToken $($graphContext.access_token) -ProvisioningPolicyGroupSearchtable $w365ProvisioningPolicyGroupSearchtable
$w365CloudPCs = Get-W365CloudPCs -AccessToken $($graphContext.access_token)

#-----#
# Retrieve Intune Total Aggregated Remote Connection Report for Windows365 #
#-----#
# Get Intune report data for Total Aggregated Remote Connections
$rawAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReport -AccessToken $($graphContext.access_token)
$enrichedAggregatedRemoteConnections = Get-W365AggregatedRemoteConnectionReportWithEnrichedData -AccessToken $($graphContext.access_token)

# Export Array of Hashtables to JSON file and CSV
ConvertTo-Json -InputObject $rawAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json"
$jsonRawData = Get-Content -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json
$jsonRawData | Export-Csv -Path "$($FileOutputPath)\Windows365RawAggregatedRemoteConnections.csv" -NoTypeInformation

# Export Array of Hashtables to JSON file and CSV
ConvertTo-Json -InputObject $enrichedAggregatedRemoteConnections -Depth 100 | Out-File -FilePath "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json"
$jsonEnrichedData = Get-Content -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.json" -Raw | ConvertFrom-Json
$jsonEnrichedData | Export-Csv -Path "$($FileOutputPath)\Windows365EnrichedAggregatedRemoteConnections.csv" -NoTypeInformation
```

Graph API PowerShell script (workflow)

```
#-----#
# FILTERING Enriched Aggregated Remote Connection Report for Windows365 #
#-----#

[int]$daysToAdd = -59
$checkDate = (Get-Date).AddDays($daysToAdd)
$dateFormatString = "MM/dd/yyyy hh:mm:ss tt"

#rule 01: No license check for PRD Cloud PCs | Exclude PRD CloudPCs as they will never expire (for now)
$filteredJsonData = $jsonEnrichedData
$filteredJsonData = $filteredJsonData.Where({ ($_.provisioningPolicyName -notlike 'PRD*') })

#rule 02: For never logged on provisioned CPCs (NeverLoggedIn = true) check CreateDate; if older than date limit, email user that
#rule 03: For logged on provisioned CPCs (NeverLoggedIn = false) check LastLoggedInDate; if older than date limit, email user that
# Filter CloudPCs with (NeverLoggedIn eq $true and CreatedDate lt $checkDate) or (NeverLoggedIn eq $false and LastActiveTime le $checkDate)
$totalFilteredJsonData = $filteredJsonData.Where({ (($_.NeverSignedIn -eq $true) -and ($_.CreatedDate -lt $checkDate) -and ($_.provisioningPolicyName -notlike 'PRD*') })

# Loop through the filtered data and add the userId to the hashtable
$enrichedTotalFilteredJsonData = $totalFilteredJsonData
$enrichedTotalFilteredJsonData | ForEach-Object {
    $userPrincipalName = $_.UserPrincipalName
    $userId = Get-W365UserId -AccessToken $($graphContext.access_token) -userPrincipalName $userPrincipalName
    $_ | Add-Member -MemberType NoteProperty -Name "UserDirectoryObjectId" -Value $userId -Force # add the userId to the hashtable
}

# Export Array of Hashtables to JSON file
ConvertTo-Json -InputObject $enrichedTotalFilteredJsonData -Depth 100 | Out-File -FilePath "C:\Temp\W365CloudPCsInactiveLast60Days.json"
$finalJsonData = Get-Content -Path "C:\Temp\W365CloudPCsInactiveLast60Days.json" -Raw | ConvertFrom-Json
$finalJsonData | Export-Csv -Path "C:\Temp\W365CloudPCsInactiveLast60Days.csv" -NoTypeInformation -Force
```

Graph API PowerShell script (workflow)

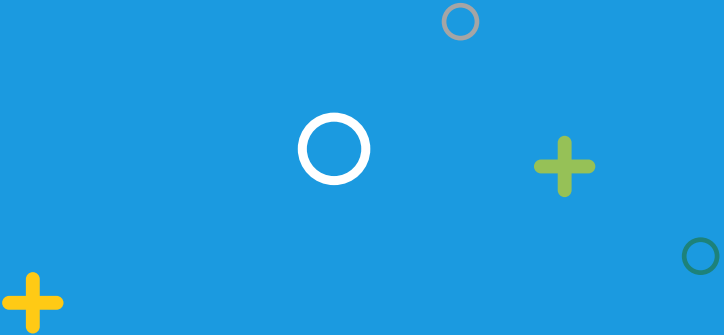
```
#-----#
# FILTERING Enriched Aggregated Remote Connection Report for Windows365 #
#-----#

[int]$daysToAdd = -59
$checkDate = (Get-Date).AddDays($daysToAdd)
$dateFormatString = "MM/dd/yyyy hh:mm:ss tt"
#rule 01: No license check for PRD Cloud PCs | Exclude PRD CloudPCs as they will never expire (for now)
$filteredJsonData = $jsonEnrichedData
$filteredJsonData = $filteredJsonData.Where({ ($_.provisioningPolicyName -notlike 'PRD*') })

#rule 02: For never logged on provisioned CPCs (NeverLoggedIn = true) check CreateDate; if older than date limit, email user that
#rule 03: For logged on provisioned CPCs (NeverLoggedIn = false) check LastLoggedInDate; if older than date limit, email user that
# Filter CloudPCs with (NeverLoggedIn eq $true and CreatedDate lt $checkDate) or (NeverLoggedIn eq $false and LastActiveTime le $checkDate)
$totalFilteredJsonData = $filteredJsonData.Where({ (($_.NeverSignedIn -eq $true) -and ($_.CreatedDate -lt $checkDate) -and ($_.provisioningPolicyName -notlike 'PRD*') })

# Loop through the filtered data and add the userId to the hashtable
$enrichedTotalFilteredJsonData = $totalFilteredJsonData
$enrichedTotalFilteredJsonData | ForEach-Object {
    $userPrincipalName = $_.UserPrincipalName
    $userId = Get-W365UserId -AccessToken $($graphContext.access_token) -userPrincipalName $userPrincipalName
    $_ | Add-Member -MemberType NoteProperty -Name "UserDirectoryObjectId" -Value $userId -Force # add the userId to the hashtable
}

# Export Array of Hashtables to JSON file
ConvertTo-Json -InputObject $enrichedTotalFilteredJsonData -Depth 100 | Out-File -FilePath "C:\Temp\W365CloudPCsInactiveLast60Days.json"
$finalJsonData = Get-Content -Path "C:\Temp\W365CloudPCsInactiveLast60Days.json" -Raw | ConvertFrom-Json
$finalJsonData | Export-Csv -Path "C:\Temp\W365CloudPCsInactiveLast60Days.csv" -NoTypeInformation -Force
```



Automated workflow



Get-W365CPC-InactiveFor60Days-Report-Data

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

Revoke-W365CPC-License

Remove User from
CloudPC license group
(Enterprise CPC)

Remove User from
CloudPC provisioning group
(Enterprise & Frontline CPC)

Notify-W365CPC-User

Send notification email
to CloudPC User

Get-W365CPC-InactiveFor60Days-Report-Data

Retrieve CloudPCReports:
getTotalAggregatedRemoteConnectionReports

Enrich report data for
readability & follow-up actions

Filter report data on
CloudPC Types &
60 days inactivity

ManagedDeviceName	DeviceType	NeverSignedIn	DaysSinceLastSignIn	contactEmail	provisioningPolicyGroupName	ExpectedEnterpriseLicenseGroupName	ExpectedEnterpriseLicenseGroupId	provisioningPolicyGroupId	UserDirectoryObjectId
TST-test0-B5JM4	Enterprise	TRUE	65	test0@cognitionit.com	W365-TST Test Cloud PC	W365-LIC-4vCPU-16GB-128GB	3da17d79-1234-1234-1234-1234567	736b9fc8-1234-1234-1234-1	ae862dc8-1234-1234-12
TST-test1-KP47C	Enterprise	FALSE		test1@cognitionit.com	W365-TST Test Cloud PC	W365-LIC-8vCPU-32GB-256GB	7bf95eb1-1234-1234-1234-1234567	736b9fc8-1234-1234-1234-1	bbb9c44d-1234-1234-12
DEV-test0-VEOL3	Enterprise	FALSE	66	test0@cognitionit.com	W365-DEV Development Cloud PC	W365-LIC-4vCPU-16GB-128GB	3da17d79-1234-1234-1234-1234567	b8483202-1234-1234-1234-1	c5a187e5-1234-1234-12
DEV-test1-ZSSX8	FrontlineDedicated	FALSE		test1@cognitionit.com	W365-DEV Development Cloud PC FL			a0249b00-1234-1234-1234-1	afb9d8b7-1234-1234-12
DEV-test2-FAUF8	Enterprise	FALSE	63	test2@cognitionit.com	W365-DEV Development Cloud PC	W365-LIC-2vCPU-8GB-128GB	e37795cc-1234-1234-1234-1234567	b8483202-1234-1234-1234-1	07e742fa-1234-1234-12

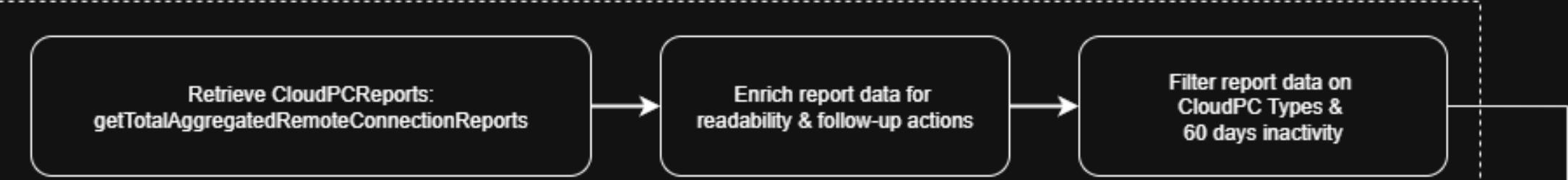
Remove User from
CloudPC license group
(Enterprise CPC)

Remove User from
CloudPC provisioning group
(Enterprise & Frontline CPC)

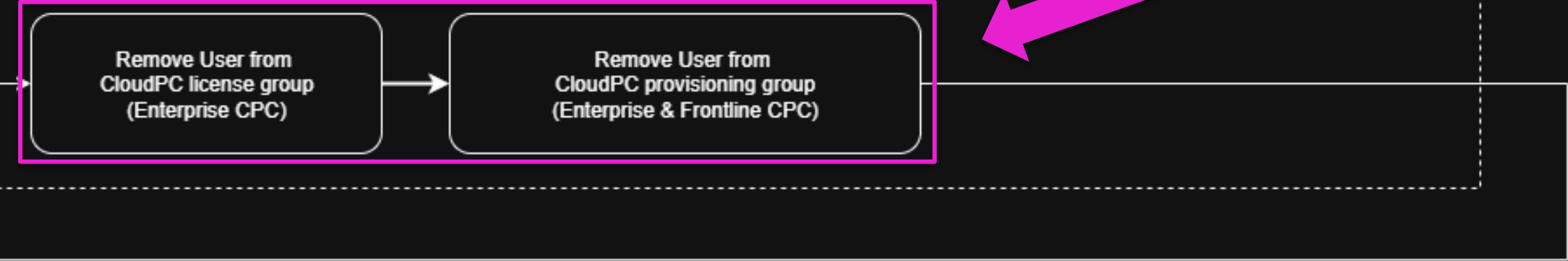
Notify-W365CPC-User

Send notification email
to CloudPC User

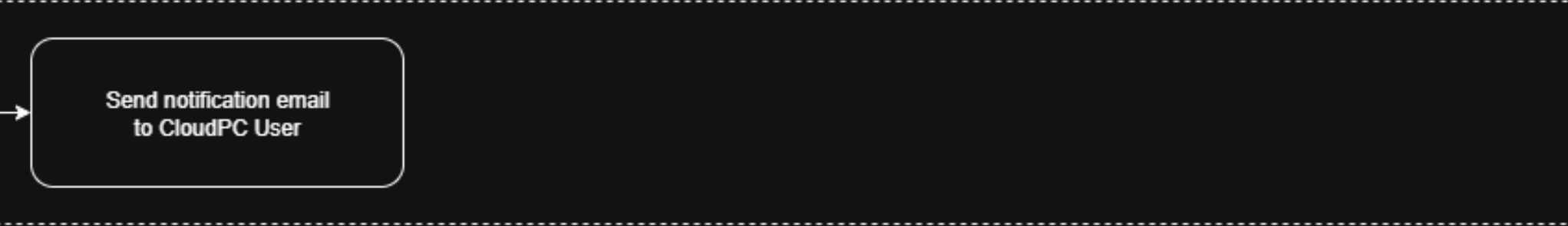
Get-W365CPC-InactiveFor60Days-Report-Data



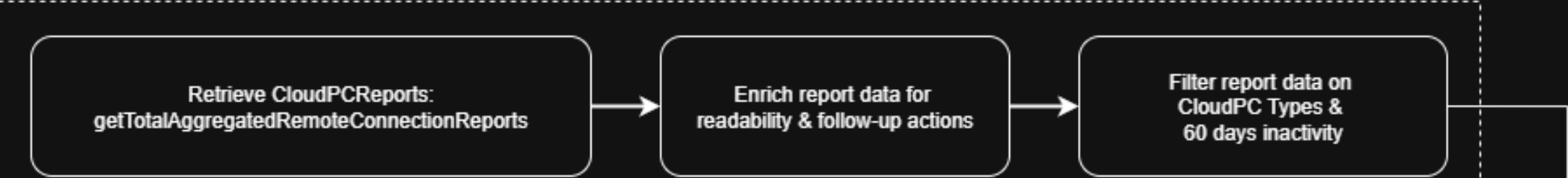
ManagedDeviceName	DeviceType	NeverSignedIn	DaysSinceLastSignIn	contactEmail	provisioningPolicyGroupName	ExpectedEnterpriseLicenseGroupName	ExpectedEnterpriseLicenseGroupId	provisioningPolicyGroupId	UserDirectoryObjectId
TST-test0-B5JM4	Enterprise	TRUE	65	test0@cognitionit.com	W365-TST Test Cloud PC	W365-LIC-4vCPU-16GB-128GB	3da17d79-1234-1234-1234-1234567	736b9fc8-1234-1234-1234-1	ae862dc8-1234-1234-12
TST-test1-KP47C	Enterprise	FALSE		test1@cognitionit.com	W365-TST Test Cloud PC	W365-LIC-8vCPU-32GB-256GB	7bf95eb1-1234-1234-1234-1234567	736b9fc8-1234-1234-1234-1	bbb9c44d-1234-1234-12
DEV-test0-VEOL3	Enterprise	FALSE	66	test0@cognitionit.com	W365-DEV Development Cloud PC	W365-LIC-4vCPU-16GB-128GB	3da17d79-1234-1234-1234-1234567	b8483202-1234-1234-1234-1	c5a187e5-1234-1234-12
DEV-test1-ZSSX8	FrontlineDedicated	FALSE		test1@cognitionit.com	W365-DEV Development Cloud PC FL			a0249b00-1234-1234-1234-1	afb9d8b7-1234-1234-12
DEV-test2-FAUF8	Enterprise	FALSE	63	test2@cognitionit.com	W365-DEV Development Cloud PC	W365-LIC-2vCPU-8GB-128GB	e37795cc-1234-1234-1234-1234567	b8483202-1234-1234-1234-1	07e742fa-1234-1234-12



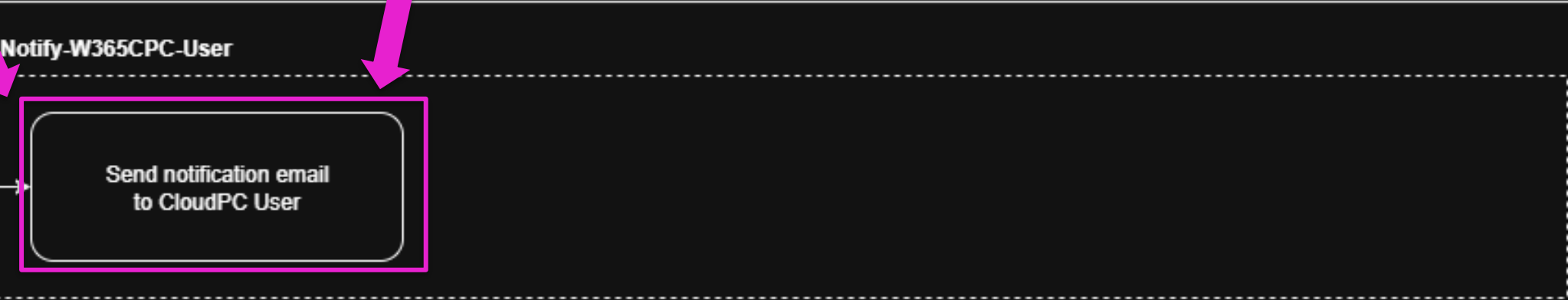
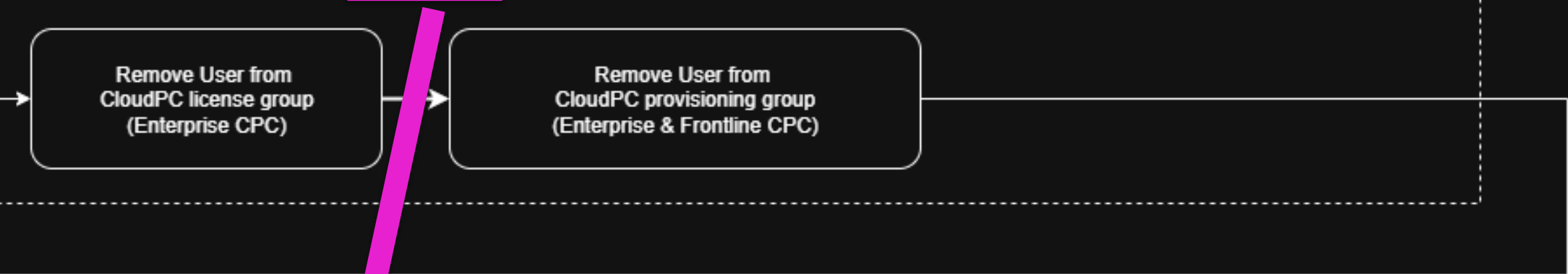
Notify-W365CPC-User



Get-W365CPC-InactiveFor60Days-Report-Data



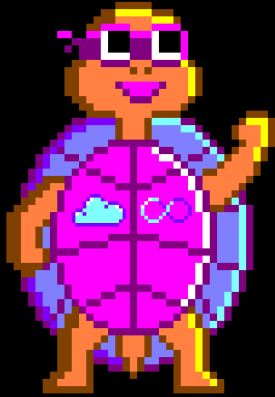
ManagedDeviceName	DeviceType	NeverSignedIn	DaysSinceLastSignIn	ContactEmail	provisioningPolicyGroupName	ExpectedEnterpriseLicenseGroupName	ExpectedEnterpriseLicenseGroupID	provisioningPolicyGroupID	UserDirectoryObjectID
TST-test0-B5JM4	Enterprise	TRUE	65	test0@cognitionit.com	W365-TST Test Cloud PC	W365-LIC-4vCPU-16GB-128GB	3da17d79-1234-1234-1234-1234567	736b9fc8-1234-1234-1234-1	ae862dc8-1234-1234-12
TST-test1-KP47C	Enterprise	FALSE		test1@cognitionit.com	W365-TST Test Cloud PC	W365-LIC-8vCPU-32GB-256GB	7bf95eb1-1234-1234-1234-1234567	736b9fc8-1234-1234-1234-1	bbb9c44d-1234-1234-12
DEV-test0-VEOL3	Enterprise	FALSE	66	test0@cognitionit.com	W365-DEV Development Cloud PC	W365-LIC-4vCPU-16GB-128GB	3da17d79-1234-1234-1234-1234567	b8483202-1234-1234-1234-1	c5a187e5-1234-1234-12
DEV-test1-ZSSX8	FrontlineDedicated	FALSE		test1@cognitionit.com	W365-DEV Development Cloud PC FL			a0249b00-1234-1234-1234-1	afb9d8b7-1234-1234-12
DEV-test2-FAUF8	Enterprise	FALSE	63	test2@cognitionit.com	W365-DEV Development Cloud PC	W365-LIC-2vCPU-8GB-128GB	e37795cc-1234-1234-1234-1234567	b8483202-1234-1234-1234-1	07e742fa-1234-1234-12





THANK YOU

Are there any questions?



☰

cloud-devops-ninja / myPresentations

🔍 Type / to search

🔗

🔄

+

🕒

🔗

📁

👤

<> Code

⦿ Issues

🔗 Pull requests

⏮ Actions

📁 Projects

📖 Wiki

🛡 Security

📈 Insights

⚙ Settings

📁 Files

🔗 main

+

🔍

🔍 Go to file

t

> 📁 Alyx

> 📁 ExpertsLive

▼ 📁 GraphAPI_StarterKit

> 📁 Code

> 📁 Postman

> 📁 Slides

▼ 📁 GraphAPI_W365LicenseManagement

▼ 📁 Code

📄 GraphAPI_W365LicenseReport.ps1

▼ 📁 Slides

📄 AVDTF_GraphAPI_W365_License_Management_v1.pdf

myPresentations / GraphAPI_W365LicenseManagement /

⋮

👤 cloud-devops-ninja

AVDTechFest update

0129856 · 4 minutes ago

🕒 History

Name	Last commit message	Last commit date
📁 ..		
📁 Code	AVDTechFest update	4 minutes ago
📁 Slides	AVDTechFest update	4 minutes ago